

PACT: Prior-Aware Coordination via Type-inference for LLM Multi-Agent Systems

Anonymous submission

Abstract

Coordinating LLM agents requires beliefs over latent goals and preferences. Prompt-based beliefs consume context and can drift, while an explicit posterior over the joint persona profile grows exponentially with the number of agents. We introduce **PACT** (Prior-Aware Coordination via Type-inference), which pairs centralized dispatch with one numeric posterior per agent. Under transition independence, reward locality, and prior factorization, these marginals exactly represent the joint posterior, reducing persona storage and update work from $O(|\Theta_i|^n)$ to $O(n|\Theta_i|)$ while retaining $\sqrt{K}$ episode dependence in posterior-sampling regret. We analyze an idealized type-discrimination bonus and evaluate a practical tensor-based proxy. In a four-backbone, environment-matched HP-SPGG control, PACT and Joint-PSRL have no resolved mean-regret gap; diagnostics show the storage reduction and a two-agent failure of the proxy. Analytic MaaSSim replays reproduce joint marginals numerically through four drivers. Constructed Concordia shows the rate pattern, corrected SOTOPIA no update gain, and separate MaaSSim ablations provide directional mechanism evidence.

1 Introduction

Large language models are increasingly deployed as autonomous agents in multi-agent settings that negotiate, bargain, allocate resources, and provide public goods (Wu et al. 2023; Li et al. 2023; Hong et al. 2024; Park et al. 2023; FAIR). In these settings each agent typically carries a private *persona*, meaning a hidden goal, preference, or constraint that determines how it values outcomes. Such preferences are rarely observable to a coordinator or to the other agents (Zhou et al. 2024; Abdulhai et al. 2023). Consider a ride-hailing platform that dispatches orders to a fleet of independent drivers. Each driver holds private operational preferences, such as avoiding long trips, staying within a familiar zone, or driving only under surge pricing. These surface only through which orders the driver accepts, declines, or reroutes. Effective coordination requires the platform to maintain and update a belief over each driver’s persona as the interaction unfolds, because the best joint assignment depends on the realized persona profile. A coordinator that recovers these preferences can route requests to drivers who will accept them. One that treats all drivers as interchangeable makes avoidable assignment errors and leaves mutually acceptable matches unrealized (Mu

et al. 2026; Yi et al. 2025).

This setting exposes two distinct inference bottlenecks. The first is representational. A common approach asks the LLM to maintain beliefs about the other agents in context (Yi et al. 2025; Mu et al. 2026). This design is flexible, but its belief state is generated text that must be interpreted and rewritten over successive turns. Errors can therefore persist or compound, and tracking quality can depend strongly on the reasoning ability of the underlying backbone (Park et al. 2024). The second bottleneck is computational. Suppose beliefs are instead stored as a numeric posterior. Let there be n agents, each drawing its persona from a library of $|\Theta_i|$ candidates. An exact distribution over the joint profile then has $|\Theta_i|^n$ entries, exponential in the number of agents. In the dispatch example the best assignment depends on the full profile rather than on any driver in isolation, so the coordinator cannot simply discard the joint structure. The two problems are separate, since replacing generated text with a numeric joint table addresses the first but not the exponential state space. A scalable method must therefore address both by moving belief maintenance outside the prompt while preserving the decision-relevant information of the joint posterior without storing it explicitly.

In this work, to address both bottlenecks, we propose **PACT** (Prior-Aware Coordination via Type-inference, Figure 1 and Algorithm 1). PACT removes belief maintenance from the prompt and keeps one numeric posterior per agent over a discrete persona library. Each episode it samples a profile from these posteriors and plans a joint policy under the sampled profile. The update uses a persona-conditioned likelihood of the observed outcome. In the finite experiments, the coordinator dispatches the selected menu action and evaluates the realized reward under an outcome scorer calibrated offline from the same backbone; HP-SPGG therefore uses an analytic Gaussian reward likelihood and no online token-probability query. In open-text SOTOPIA-Hard, the LLM generates utterances but the retained adapter uses an explicitly approximate intent-menu surrogate. Thus the experiments do not rely on next-token likelihoods. Because the belief state is numeric rather than generated text, it is carried forward and reused for planning instead of being rewritten in context. The theory below provides two complementary guarantees: Posterior Decoupling establishes when the marginals exactly represent the full joint posterior, and the Bayesian-regret theorem controls

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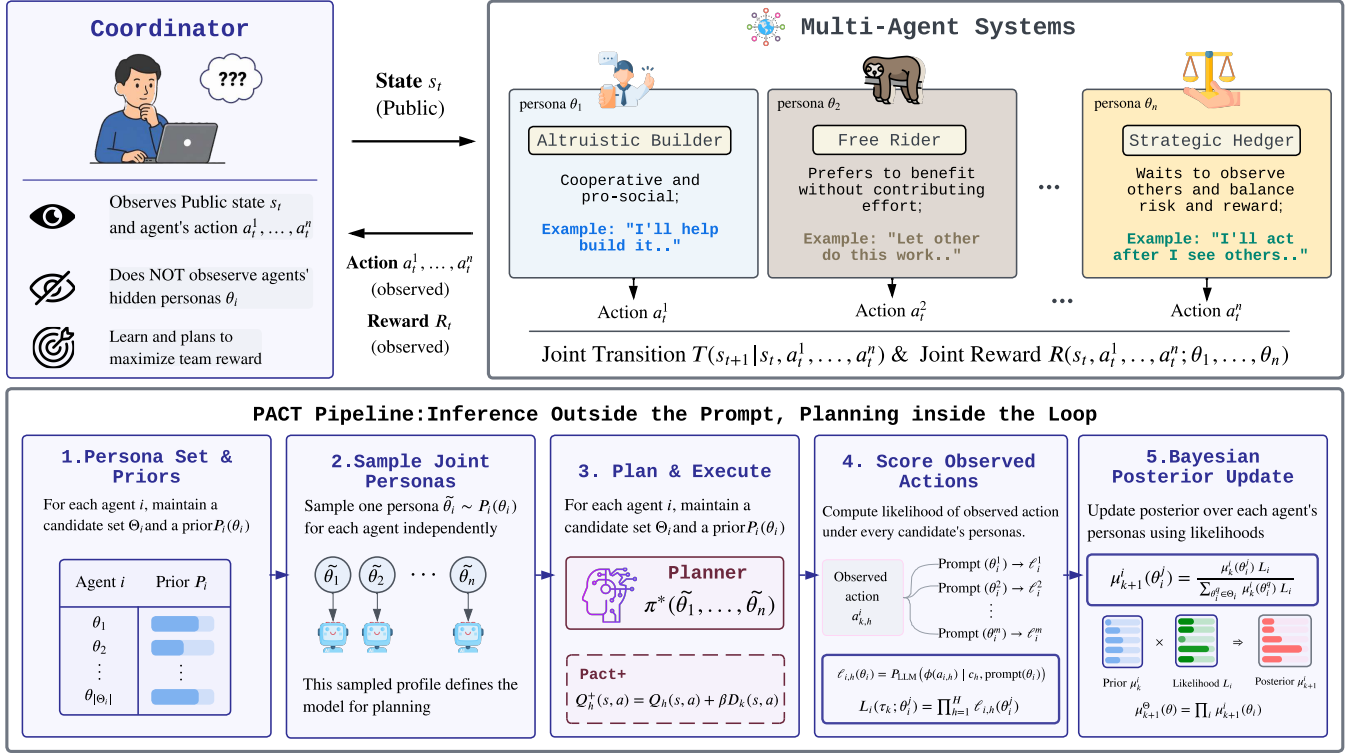


Figure 1: **Overview of PACT.** **Top:** A coordinator interacts with LLM agents in a public environment; each agent has a private persona (e.g., altruistic builder, free rider, strategic hedger) hidden from the coordinator. **Bottom:** (1) per-agent persona prior; (2) sample a joint profile from per-agent posteriors; (3) plan and execute (PACT⁺ adds a type-discrimination bonus); (4) score each observed outcome under every candidate persona’s calibrated outcome scorer; (5) Bayes-update each posterior. Under TI, RL, PF: per-agent updates equal the full joint posterior at linear-in- n storage.

the decision loss of learning those beliefs.

Our contributions are threefold. First, PACT replaces recurrent in-prompt belief rewriting with an external numeric posterior and a centralized planner. The finite implementations use an offline persona-conditioned outcome scorer; the open-text adapter is explicitly treated as a surrogate. We also study an idealized pairwise-disagreement exploration bonus and evaluate a practical variance-based proxy in PACT⁺. Second, under transition independence, reward locality, and prior factorization, Posterior Decoupling proves that the joint persona posterior factors into per-agent marginals, reducing persona storage and update work from $O(|\Theta_i|^n)$ to $O(n|\Theta_i|)$. PACT then inherits posterior-sampling regret with $\sqrt{K}$ dependence in the episode count, while retaining explicit dependence on the horizon, state space, joint action space, persona library, and planner error (Theorem 3.3). Third, we separate aligned diagnostics from transfer boundaries. A four-backbone HP-SPGG control finds no resolved PACT–Joint-PSRL mean-regret gap and places the practical PACT⁺ proxy lowest in the matched $K=20$ cells. Separate scaling experiments demonstrate the storage growth but include a two-agent bonus failure that we report explicitly. A constructed iterated Concordia diagnostic reproduces the plateau-versus-linear pattern under imposed structure. In open-text SOTOPIA-Hard, a corrected tracker updates but yields no score gain.

2 Related Work

Bayesian and latent-type RL foundations. Posterior-sampling RL provides near-optimal single-agent Bayesian regret guarantees (Strens 2000; Osband, Russo, and Van Roy 2013; Agrawal and Jia 2017; Ouyang et al. 2017), with information-theoretic analyses in Russo and Van Roy (2014, 2016); Dasgupta et al. (2025). Markov-game results mainly cover known-type or zero-sum settings (Jahromi, Jain, and Nayyar 2024; Xiong et al. 2022), and latent MDPs are exponentially hard without identifiability (Kwon et al. 2021; Nguyen-Tang and Arora 2024). Harsanyi-Bellman Ad-hoc Coordination (Albrecht and Ramamoorthy 2015, 2019) is closely related in its use of policy types, but it does not provide the finite-time regret result studied here. Structured Bayesian RL with causal priors (Mutti et al. 2024) and LLM-based PSRL (Arumugam and Griffiths 2025) are related but do not exploit factorization across agent personas. Work on general-sum Markov-game learning (Jin et al. 2024; Song, Mei, and Bai 2021; Mao and Başar 2023; Erez et al. 2023; Yang et al. 2025) and PSRO-style population learning (Lanctot et al. 2017; Muller et al. 2019; Marris et al. 2021) is complementary to our finite centralized planning step.

LLM-driven coordination. Recent LLM-agent methods often perform persona inference through prompt-based belief reasoning. ECON (Yi et al. 2025) represents beliefs through LLM-generated hypotheses, while A-ToM (Mu et al. 2026)

uses theory-of-mind layers combined with online expert aggregation (Cesa-Bianchi et al. 1997). These methods are related to interactive-POMDP formulations (Gmytrasiewicz and Doshi 2005), but their belief state remains part of the model context. PACT instead keeps the belief state as an explicit posterior and uses persona-conditioned action probabilities to update it.

LLM multi-agent systems and dialogue agents. Orchestration frameworks (Wu et al. 2023; Li et al. 2023; Hong et al. 2024; Park et al. 2023; Chen et al. 2023) and negotiation systems (FAIR; Abdulhai et al. 2023) treat partner modeling as a side-effect of role prompting or a learned reward (Agashe et al. 2025). PACT specifies the inference step these orchestrators could use when partner personas are discrete and prompt-addressable, evaluated on HP-SPGG, Concordia (Vezhnevets et al. 2023; Smith et al. 2026), and SOTOPIA (Zhou et al. 2024).

3 Methodology

3.1 Preliminaries and Definitions

We formulate LLM multi-agent coordination as a Bayesian inference problem. A coordinator interacts with n LLM agents for K episodes of H turns each. Each agent i is assigned a private persona $\theta_i^* \in \Theta_i$ through its system prompt. The joint persona profile $\theta^* \in \Theta = \prod_i \Theta_i$ is drawn once from the prior P_0^Θ and held fixed. At each turn, the coordinator plans from its current posterior and the public state. In tabular substrates, execution dispatches a menu action, whereas in LLM substrates it queries each hidden-persona agent under the current public context and records the emitted action token. The coordinator observes these actions, the per-agent rewards $r_i(s, a, \theta_i^*) \in [0, 1]$, and the next state generated by an unknown transition kernel P^* . Both P^* and θ^* are inferred from trajectories. Planning is represented by a fixed deterministic centralized map $\mathcal{P}$ from a sampled world to a joint policy. In the finite experimental substrates, $\mathcal{P}$ uses exhaustive joint-action argmax with deterministic tie-breaking. SOTOPIA-Hard uses direct open-text generation and has no finite planner. Performance is measured by cumulative Bayesian regret relative to the output of the same planner under full knowledge of (P^*, θ^*) . Full notation and formal definitions are deferred to Appendix B.2.

3.2 PACT

PACT keeps one numeric posterior $\mu_k^{\Theta, i}$ per agent over the persona library and a model posterior μ_k^P , coupled to the centralized planner $\mathcal{P}$ (Algorithm 1). Each episode it samples a world $(\hat{P}_k, \hat{\theta}_k)$ and calls $\mathcal{P}$ for a joint policy. HP-SPGG uses backward induction with exhaustive joint-action argmax, and compact Concordia uses one-step exhaustive enumeration. SOTOPIA-Hard instead generates open-text utterances and has no finite planner. After observing episode k , each agent’s numeric belief updates independently as

$$\mu_{k+1}^{\Theta, i}(\theta_i) \propto \mu_k^{\Theta, i}(\theta_i) \prod_{h=1}^H q_\phi(o_{i,h}^k | x_{i,h}^k, \theta_i) \quad (1)$$

Algorithm 1 PACT (idealized PACT⁺ uses Eq. 2; experiments use the disclosed proxy)

Input: priors $P_0^P, \{P_0^{\Theta, i}\}$; horizon H ; episodes K ; deterministic centralized planner $\mathcal{P}$; bonus coefficient $\beta \geq 0$.

- 1: Initialize $\mu_1^P \leftarrow P_0^P, \mu_1^{\Theta, i} \leftarrow P_0^{\Theta, i}$ for all i .
- 2: **for** $k = 1, \dots, K$ **do**
- 3: Sample $\hat{P}_k \sim \mu_k^P$ and $\hat{\theta}_k^i \sim \mu_k^{\Theta, i}$ independently.
- 4: If $\beta > 0$: compute $D_k(s, a)$ via Eq. (2) and add βD_k to Q in backward induction.
- 5: $\pi_k \leftarrow \mathcal{P}(\hat{P}_k, \hat{\theta}_k)$; execute π_k , observe τ_k .
- 6: Bayes-update each marginal independently from τ_k .
- 7: **end for**

where $x_{i,h}^k$ is the public context and $o_{i,h}^k$ the observed action and reward. Calibration elicits one numeric score per persona-by-profile cell from the backbone, and the resulting tensor serves as both the outcome scorer of the planner and the mean of the tracker’s likelihood. In HP-SPGG, q_ϕ uses the calibrated Gaussian reward factor $\exp[-(r_{i,h} - \hat{r}_i(s_h, a_h, \theta_i))^2 / (2\sigma^2)]$ with $\sigma=0.08$; its action factor is uninformative, so the tracker issues no model queries online.

The factored belief stores $O(n|\Theta_i|)$ entries, whereas the naive joint posterior $\mu_k(\theta^*)$ has $|\Theta| = \prod_i |\Theta_i|$, exponential in n . However, it is exact only for a specific class of coordination problems, and identifying that class is part of our contribution. Three structural conditions are jointly sufficient, and (TI) and (RL) are individually necessary for the likelihood to split across agents (Appendix B.4).

Assumption 3.1 (Structural conditions). **(TI) Transition independence.** *Environment dynamics do not depend on the persona profile: $P^*(s' | s, a, \theta) = P^*(s' | s, a)$.* **(RL) Reward locality.** *Each agent’s reward depends only on its own persona: $r_i(s, a, \theta) = r_i(s, a, \theta_i)$.* **(PF) Prior factorization.** *The coordinator’s prior over personas factors across agents: $P_0(\theta^*) = \prod_i P_0^{\Theta, i}(\theta_i^*)$.*

(TI), (RL), and (PF) are explicit applicability conditions rather than assumptions about all social LLM interaction. Specifically, (TI) holds when the transition rule is persona-blind. (RL) excludes cross-agent utilities such as envy or spite, $r_1 = r_1^{\theta_1} - \alpha r_2^{\theta_2}$, which break the factored update (Appendix B.4). (PF) is met when slots are filled independently from a persona library. Together they define the class for which posterior decoupling is exact. Section 4.3 extends the evaluation to open-ended dialogue via an intent-menu surrogate.

Theorem 3.2 (Posterior Decoupling). *Under Assumption 3.1, the joint posterior over (P^*, θ^*) at every episode k factors exactly into*

$$P(P, \theta | \mathcal{H}_k) = \mu_k^P(P) \cdot \prod_{i=1}^n \mu_k^{\Theta, i}(\theta_i),$$

where each persona marginal $\mu_k^{\Theta, i}$ follows the per-agent Bayes update in Eq. (1). Posterior storage is $O(n|\Theta_i|)$ rather than $O(|\Theta|^n)$.

The proof factors the trajectory likelihood across agents using (TI) and (RL) and closes by induction on k under (PF). The full argument is in Appendix B.3. Consequently, each persona-belief update evaluates $O(n|\Theta_i|)$ candidate likelihoods rather than $O(|\Theta_i|^n)$ joint profiles. This resolves the computational bottleneck in persona-posterior representation and updating.

Complexity. Per episode, PACT performs one planner call over the joint action menu and $nH|\Theta_i|$ closed-form likelihood evaluations, and it stores $O(n|\Theta_i|)$ persona parameters beside the model posterior. Decoupling therefore makes belief storage and updating linear in n . The prompt-based comparators instead issue $n \cdot H$ action calls per episode and carry the belief in context. Appendix C.1 reports measured call and token costs.

3.3 PACT⁺: Type-Discrimination Bonus

PACT may learn slowly when the coordinated policy repeatedly visits states where different candidate personas give the same reward. In this case the observations provide little information about type, and the posterior may remain close to the prior. We formalize this case as the *deep-trap* family in Appendix B.7. To mitigate this, PACT⁺ adds a small planning bonus at each (s, a) where candidate personas differ under the planner’s outcome scorer,

$$D_k(s, a) := \sum_{i=1}^n \mathbb{E}_{\theta, \theta' \sim \mu_k^{\Theta_i, i}} [r_i(s, a, \theta) - r_i(s, a, \theta')], \quad (2)$$

where θ and θ' are drawn independently from $\mu_k^{\Theta_i, i}$. This idealized bonus costs $O(n|\Theta_i|^2)$ per state-action pair and vanishes as the posterior concentrates; the formal PACT⁺ results below apply to Eq. (2). The reported experiments use a cheaper tensor-based proxy, $\sum_i (1 - \max_{\theta} \mu_k^{\Theta_i, i}(\theta)) \text{Var}_{\theta} [r_i(s, a, \theta)]$, plus 0.05 times the action-profile spread, exactly as implemented in the released coordinator. That final spread term need not vanish, so the empirical proxy is not claimed to inherit the idealized bonus theorem. We use $\beta = 0.25$, selected on disjoint HP-SPGG seeds and then frozen; $\beta = 0$ recovers PACT.

Theoretical guarantees. The theory answers two separate questions. Theorem 3.2 establishes that the compact factored posterior is exact and removes the exponential persona-belief representation. The theorem below establishes the statistical price of learning and acting from that posterior.

Theorem 3.3 (Per-agent Bayesian regret of PACT). *Let initial states be drawn independently of the model and of the sampling randomness, let the coordinator use an ϵ_{plan} -approximation of a fixed measurable deterministic planner $\mathcal{P}$, and place a Dirichlet prior on P^* . Under Assumption 3.1, for every agent i ,*

$$\mathbb{E}[\text{Reg}_i^B(K)] \leq \tilde{O}(H^{3/2} \sqrt{SA_{\text{joint}}K}) + O(H \sqrt{|\Theta_i| \mathcal{H}(P_0^{\Theta_i, i})K}) + K\epsilon_{\text{plan}}, \quad (3)$$

where $A_{\text{joint}} = \prod_i |A_i|$ and $\mathcal{H}(P_0^{\Theta_i, i})$ is the entropy of agent i ’s persona prior. The finite HP-SPGG and compact Concordia implementations enumerate their action menus

exactly, so $\epsilon_{\text{plan}} = 0$; the theorem is not claimed for open-text SOTOPIA-Hard. Due to space constraints, the full statement and proof are deferred to Theorem B.12 in Appendix B.

For fixed problem dimensions and an exact planner, both channels scale as $\sqrt{K}$ in the episode count. The persona term depends on the per-agent library and prior entropy rather than the joint persona space, while the transition term retains $\sqrt{SA_{\text{joint}}K}$; decoupling therefore reduces persona representation and updating, not centralized planning complexity. Under a type-discrimination margin ρ , the persona term collapses to a K -independent constant (Proposition B.16). There is also an instance in the structural class on which PACT has $\tilde{O}(\sqrt{K})$ regret while type-agnostic sampling incurs $\Omega(K)$ (Theorem B.18). On the constructed deep-trap family, the idealized bonus of Eq. (2) reduces $\Omega(|\Theta_i|^n)$ burn-in to $O(1)$ (Theorem B.20); this theorem is not asserted for the empirical variance-and-spread proxy. Under coupled RNG and the structural assumptions, PACT is pathwise identical to Joint-PSRL (Proposition B.9). Finally, a separate counterexample shows exponential burn-in is possible once reward locality is removed, but it does not isolate prior factorization or planning complexity (Theorem B.21).

4 Numerical Evaluation

The numerical evaluation follows three questions that mirror the representation and computation bottlenecks of Section 1, then attribute the observed differences to individual components.

- **RQ1 (representation):** How does an out-of-prompt numeric posterior compare with prompt-maintained beliefs under aligned finite structure, and where does that comparison stop transferring?
- **RQ2 (computation):** Does factored tracking reproduce explicit-joint inference while reducing persona storage and update work as n grows?
- **RQ3 (attribution):** What evidence separates the closed-form update, identity attachment, discrimination bonus, and centralized dispatch?

Each substrate answers only the questions its design permits. HP-SPGG (Section 4.1) imposes TI/RL/PF and provides the environment-matched control. Concordia (Section 4.2) uses externally authored exact payoffs, one-shot for RQ1 and iterated for RQ2/RQ3. MaaSSim (Section 4.4) tests all three through fixed-state and regenerated-market replays. SOTOPIA-Hard, where the assumptions cannot be verified, marks the RQ1 transfer boundary. The substrates are therefore diagnostics rather than interchangeable leaderboards. The benchmark evaluations use four LLM backbones (DeepSeek-V3.2, GPT-5.4-nano, Kimi-K2.6, Llama-4-Maverick-17B-128E-Instruct-FP8), while the case study uses a single pinned backbone. Each substrate reports its native metric (cumulative regret on HP-SPGG, focal payoff or score on Concordia and SOTOPIA-Hard, dispatch utility on MaaSSim).

4.1 HP-SPGG: Controlled Diagnostic Benchmark

Setup. HP-SPGG (heterogeneous-persona public-goods game) is an iterated finite public-goods simulator with $n = 3$

player slots and $K = 20$ episodes. Persona-conditioned payoffs are elicited once from each backbone and pinned as the reward tensor; the online PACT-family trajectories then use deterministic menu actions and tensor rewards rather than querying autonomous player LLMs. Each hidden type is drawn once, uniformly and independently, from $|\Theta_i| = 4$ persona archetypes spanning the taxonomy of Fischbacher, Gächter, and Fehr (2001), so TI, RL, and PF hold by construction. PF prior and DGP probes are in Appendix A.3, and the full specification is in Appendix A.1.

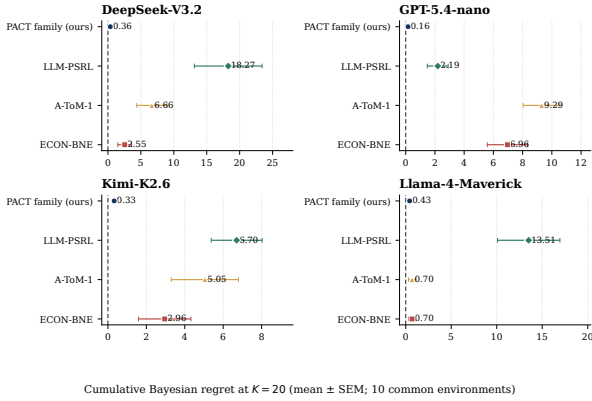


Figure 2: Environment-matched HP-SPGG cumulative Bayesian regret at $K=20$ (mean $\pm$ SEM, 10 common seeds). Figures 2, 4, and 6 use the same displayed systems, order, colors, and markers: PACT family (practical PACT⁺ here), LLM-PSRL, A-ToM-1, and ECON-BNE; the dashed zero is the oracle reference. Methods share type profiles and the pinned tensor but generate their own histories. PACT-family planning sees the complete tensor; prompted comparators see realized histories.

Baselines. The matched control compares Oracle, PACT⁺, PACT, Joint-PSRL, LLM-PSRL, PSRL-NoType, A-ToM-0/1/2, and ECON-BNE. Full definitions and hyperparameters are in Appendix A.1.

RQ1: numeric versus in-prompt systems. Figure 2 reports the fresh control. PACT⁺ has the lowest mean among the four displayed systems on every backbone: 0.360 ± 0.142 (DeepSeek), 0.159 ± 0.038 (GPT-5.4-nano), 0.325 ± 0.088 (Kimi), and 0.430 ± 0.217 (Llama), with SEM over ten common environments. Relative to the lower-regret displayed ECON-BNE/A-ToM-1 comparator, the descriptive ratio is $1.63\times\text{--}43.77\times$; the paired Llama gap is not resolved at a two-sided 95% level. The full control additionally contains structural baselines reported in the appendix. Only PACT-family and Joint-PSRL planners use the complete pinned tensor, whereas prompted comparators observe their own realized histories, so this is a system-level comparison rather than an isolation of numeric storage. The zero-online-call property is likewise specific to this finite tensor replay (Appendices A.1 and C.1).

RQ2: factored versus explicit joint. In the fresh $n=3$ control, paired PACT-minus-Joint-PSRL gaps are $0.030 \pm$

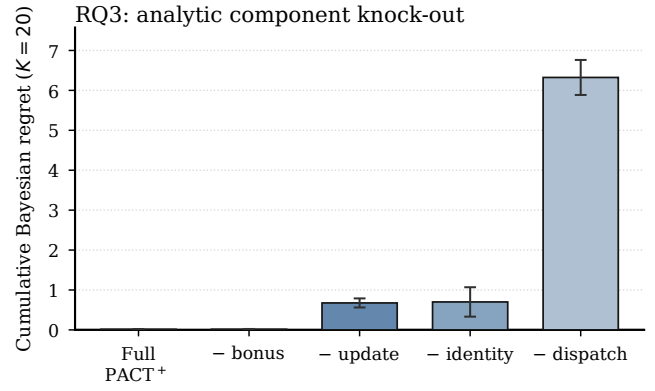


Figure 3: E-G analytic HP-SPGG component ladder ($n=3$, $|\Theta_i|=4$, $K=20$, $\beta=0.25$, 10 common environments; mean $\pm$ SEM). Full PACT⁺ has 0.015 ± 0.007 regret. The bonus knock-out is unresolved (0.016 ± 0.007); removing update gives 0.675 ± 0.114 , shuffled identity 0.700 ± 0.368 , and decentralized dispatch 6.324 ± 0.439 . Paired 95% CIs exclude zero only for update and dispatch; the identity-minus-update CI also crosses zero. Oracle regret is the black zero line.

0.294 , 0.153 ± 0.175 , -0.095 ± 0.180 , and -0.074 ± 0.346 across DeepSeek, GPT, Kimi, and Llama (mean $\pm$ SEM); none is resolved with ten seeds. The scaling diagnostic reaches a $51\times$ joint-to-factored storage ratio at $n = 5$ without a resolved PACT-Joint gap (Figure 8). This is finite-sample support, not a pathwise test of Theorem 3.2. Under a coupled prior-violating DGP, PACT remains within $1.52\times$ of the correctly supported joint baseline and is lower-regret in 9 of 15 cells (Appendix A.3).

RQ3: unified component ladder. E-G holds one analytic HP-SPGG tensor, oracle, prior, and ten environment seeds fixed while removing one operational component at a time (Figure 3). Removing the bonus is unresolved ($1.05\times$ Full), so the preregistered $1.5\text{--}3\times$ bonus cost does not appear on this kernel. Removing updating raises mean regret $45.6\times$ with a paired 95% CI excluding zero. Shuffling identity raises the mean $47.3\times$, but a few high-regret seeds leave its paired CI crossing zero and it is indistinguishable from the no-update control. Removing shared sampling and joint dispatch is the largest wall: $427\times$ Full with a paired CI excluding zero. Thus update and dispatch carry the resolved effects here; identity has the expected mean direction but not resolved significance (Appendix A.5).

4.2 Concordia-Derived Compact Benchmarks

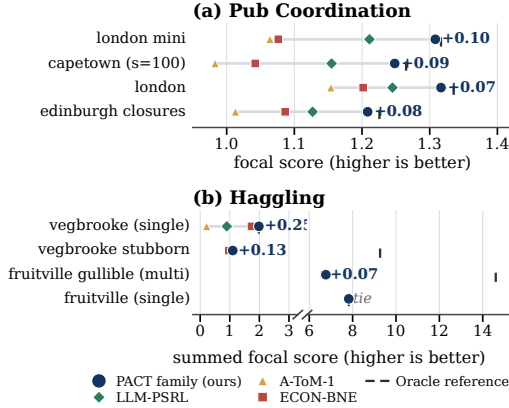


Figure 4: Concordia focal scores for the same four displayed systems as Figure 2; PACT family denotes practical PACT⁺. (a) Pub uses `oracle_joint`; (b) Haggling uses `exact_oracle_focal`. Navy labels show the PACT-family margin over the best available displayed comparator; absent points were not retained for that configuration. Full method grids are in Appendix Figures 23 and 24.

We next evaluate on the public Concordia substrates (Vezhnevets et al. 2023; Smith et al. 2026). *Pub Coordination* models agents choosing among pubs with payoff depending on shared-venue coordination, and *Haggling* models buyer-seller negotiation over price and bundles. We evaluate 9 configurations per substrate (Figure 4) and report focal-agent payoff, since LLM coordinators typically represent one party rather than a social planner. The evaluation uses a compact mechanistic evaluator that retains the official configurations’ static case parameters without executing their scripted NPC policies. Each scene is therefore a one-shot, fully enumerated decision over the analytic `payoff_for_case` function, with PF imposed rather than tested (Appendix A.6). The payoff function, action menus, and case parameters are externally sourced rather than our own design.

RQ1: one-shot decision quality. Figure 4 shows the largest decision-relevant margins; all 18 configurations appear in Appendix Figure 23. On Pub Coordination, PACT⁺ stays within 1% of `oracle_joint`. On Haggling it wins or ties every non-oracle method on all nine configurations and reaches the exact focal ceiling on five. Four transfer-price cases retain headroom up to 14.60 because `oracle_joint` optimizes total rather than focal payoff; Appendix Figure 24 exposes that objective mismatch. These are exact-payoff one-shot decisions, not recurrent-inference evidence.

RQ2 and RQ3: iterated diagnostic. A fixed-persona $K = 20$ variant uses six configurations selected on seeds 0–4 and disjoint report seeds 1000–1004. Figure 5(a) gives PACT-minus-Joint-PSRL -0.017 ± 0.044 SEM at $K = 20$; all 20 Student- t intervals cover zero. Panel (b) gives PACT⁺ regret 0.352 ± 0.035 and late rate 0.0020 ± 0.0017 , versus 1.361 ± 0.172 and 0.0764 ± 0.0139 for PSRL-NoType. Its update knock-out is directionally consistent with the

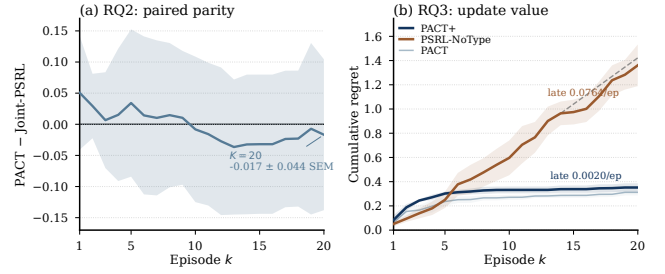


Figure 5: Iterated Concordia diagnostic ($K=20$, six held-out configurations, five report seeds, exact-payoff backbone-invariant replay with imposed PF). (a) RQ2: PACT-minus-Joint-PSRL paired mean; the two-sided Student- t 95% band over seeds covers zero at every episode. (b) RQ3: PACT⁺ plateaus while PSRL-NoType accumulates regret (late rates 0.0020 vs. 0.0764/episode; seed-level SEM); thin PACT is a reference.

resolved E-G update effect, although the substrates and ratios differ. The replay is backbone-invariant with externally authored payoffs and imposed PF. Complete per-(method, config, seed, episode) data and aggregation rules accompany the artifact.

4.3 SOTOPIA-Hard: Open-Ended Dialogue Evaluation

SOTOPIA-Hard (Zhou et al. 2024) is a dyadic, open-text, judge-scored boundary test whose curated cases do not satisfy a tested factored prior. Its adapter uses a keyword-weighted intent surrogate rather than a calibrated likelihood. Historical methods fall within a 0.12 score envelope, but the retained tracker was miswired and never updated, so those rows are descriptive only. A corrected GPT-nano rerun updates recurrently—proxy mass reaches 0.304 on *revenge_plot*—yet exceeds no corrected control. We therefore make no update-gain claim and use SOTOPIA only to mark the open-text boundary (Appendix A.4).

4.4 Case Study: Ride-Hailing Dispatch on MaaSSim

MaaSSim (Kucharski and Cats 2022) instantiates the dispatch example with hidden driver rules revealed through routed offers. The five-seed LLM suite is directional: a *score-assisted PACT hybrid* receives numeric assignment scores unavailable to pure prompt baselines, and the original paired seed rows were not retained.

RQ1: margin under designed conflict. Holding the reject penalty at 5, the measured $\lambda \in \{0, 0.5, 1\}$ replay moves from retained reject-stress offers to full persona-risky offers. The hybrid’s mean advantage over the best displayed prompt comparator grows across that sweep while its oracle-regret mean stays within 2.7–4.4 units (Figure 6(a,b)). Because the hybrid has privileged assignment scores, this localizes a decision-information margin but is not a pure belief-representation comparison.

For RQ2, E-E regenerates self-consistent Nootdorp markets for $n \in \{2, 3, 4, 6, 8\}$ and $\lambda \in \{0, 0.5, 1\}$, then runs explicit-joint tracking through $n = 4$ on the factored tracker’s saved

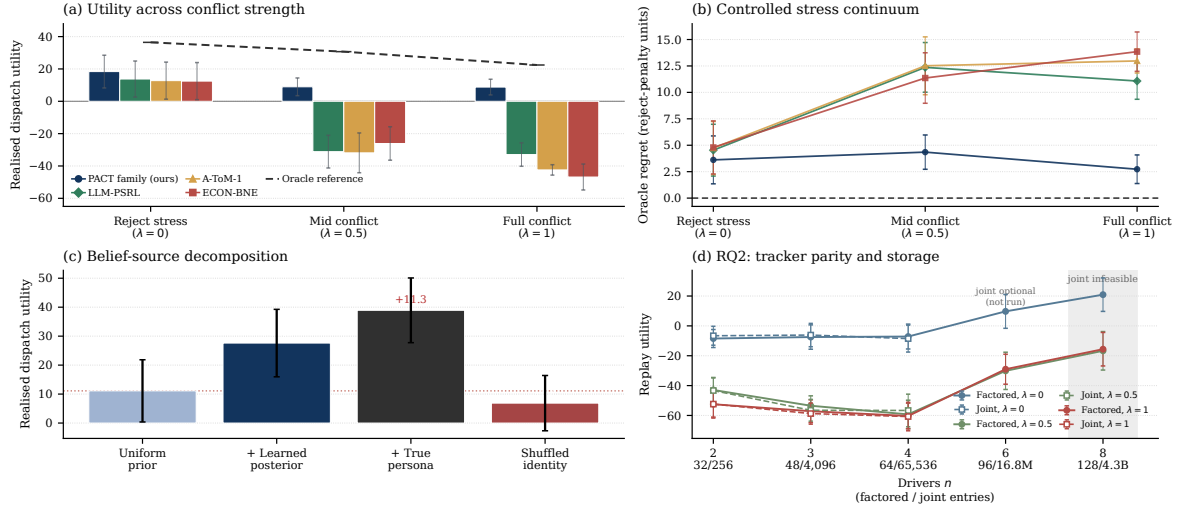


Figure 6: MaaSSim RQ diagnostics. (a,b; RQ1) The same four displayed systems as Figures 2 and 4, with PACT family denoting the score-assisted hybrid, plus a neutral oracle reference; utility and oracle regret span a designed conflict continuum (5 seeds; reject penalty 5). (c; RQ3) Learned, prior, oracle, and shuffled-identity beliefs on common states (10 seeds). (d; RQ2) Factored and explicit-joint trackers on regenerated sub-fleets at $\lambda \in \{0, 0.5, 1\}$ (10 common environment indices), using identical events but independent samples. Maximum marginal TV is 2.55×10^{-14} ; eight of nine nominal utility CIs cover zero, with the $n=2, \lambda=0$ exception retained (Appendix A.5). Error bars: SEM in (a,c,d), propagated SEM in (b).

events. Maximum marginal TV is 2.55×10^{-14} and eight of nine nominal utility CIs cover zero. The retained exception is $n = 2, \lambda = 0$: +1.85 joint-minus-factored utility, 95% CI [0.19, 3.51]. With identical marginals and planner, only the intentionally independent sample paths differ. At $n = 4$, storage is 64 versus 65,536 entries and update time is $9.03 \mu\text{s}$ versus 0.55 ms (Appendix Figure 7).

For RQ3, belief-source replay gains +16.5 mean utility over a uniform prior, while shuffled identities fall below it. Rule accuracy reaches 0.73 and exact-type mass 0.26 within eight observations; rejection events are more informative than acceptances (0.038 vs. 0.022 accuracy gain). The frozen $\beta = 0.25$ bonus changes only 4 of 406 assignments and has paired PACT⁺-minus-PACT utility -0.247 with 95% CI $[-0.806, 0.312]$. The shuffled and dispatch directions match the corresponding E-G knock-outs, but E-G resolves only dispatch and MaaSSim lacks paired LLM-suite rows. Together these results triangulate update, identity, bonus, and centralization without factorially isolating every cross-substrate interaction (Appendix A.5).

5 Conclusion

Under TI/RL/PF, PACT’s per-agent marginals exactly represent the joint persona posterior with $O(n|\Theta_i|)$ rather than $O(|\Theta_i|^n)$ storage and update work; the regret bound retains joint-planning factors. HP-SPGG has no resolved PACT-Joint gap at $n = 3$, and MaaSSim reproduces joint marginals numerically through $n = 4$, while retaining one independent-sampling utility non-parity. Iterated Concordia supports the rate pattern, corrected SOTOPIA yields no update gain, and MaaSSim’s component evidence remains directional. The guarantees therefore cover a narrow finite class, not open-ended coordination generally.

Limitations

PACT assumes fixed personas, finite actions, and an offline scorer; drift and misspecification remain open, and TI/RL/PF are only sufficient. The RL stress couples beliefs without a decision gap, while the lower bound also confounds action-space growth (Appendix A.3; Theorem B.21). SOTOPIA uses a surrogate and Concordia is one-shot. MaaSSim E-E tests representation, not performance; PACT⁺ is a proxy and joint planning remains exponential.

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A Claim-Organized Empirical Supplement

A.0 Reading Guide

This subsection certifies no standalone result; it maps each of the six paper claims to its complete evidence bundle and location. Every empirical figure and table retained below appears in one of these rows, with the single exception of Appendix A.6, whose figures and table document the oracle references and the one-shot decision-quality comparison as shared evaluation infrastructure rather than certifying a claim of their own.

Table 1: Claim-to-evidence reading guide. The appendix is organized by claim rather than substrate or experiment identifier.

Claim	Evidence	Location
C1	Exact- Table 2; Figures 9, 10, 8, 7; Tables 4, 3. ness/storage	§A.1
C2	Rate sep- Figures 11(a), 12, 5; Table 5. aration	§A.2
C3	Ideal- Theorem B.20; Figure 11(b); Figures 13, 14, 3; sized burn-in Tables 6, 12. / proxy behavior	§B.7; §A.2; §A.5
C4	Struc- Figure 15; Table 7; Figure 16; Table 8. tural class	§A.3
C5	Open- Tables 9, 10; Figures 17, 18; Table 11. text bound- ary	§A.4
C6	Mecha- Figures 3, 19, 20, 21, 22; Tables 12, 13, 14, 15. nism	§A.5

A.1 Exactness and Storage (C1)

This subsection certifies C1: under TI/RL/PF, the factored persona posterior matches the joint representation while reducing persona storage from $O(|\Theta_i|^n)$ to $O(n|\Theta_i|)$. The fresh matched control anchors the comparison on common environments and pinned tensors; the scaling figures and storage table isolate representation size from planning and likelihood differences.

Matched-likelihood source audit. The historical source audit found algorithm-specific seeds ($120000 + 10000j + s$) and no retained c19 tensors or response caches, so those unmatched rows are provenance only and are excluded from Figure 2 and Table 2. The anonymous release records this source under an anonymized provenance identifier.

Fresh environment-matched control. For each backbone we generated and pinned a complete 125-profile $\times$ 3-player $\times$ 4-persona live outcome-score tensor (1,500 judge cells), then ran all methods over the same ten environment seeds, type profiles, uniform prior, and exact exhaustive tensor oracle with $K=20$, $\beta=0.25$, and Gaussian observation scale 0.08. There is no additional stochastic board state. Each method receives its own realized-reward history generated from the shared tensor. The strongest fixed LLM-coordination comparator is selected by minimum mean regret within ECON-BNE and A-ToM-0/1/2. Ratios divide per-backbone means; paired

gaps use common seed indices. Provider sampling seeds are unavailable, so this is environment matching rather than pathwise provider-RNG matching; every accepted raw response is content-hash cache-pinned. The same pinned tensor defines the HP-SPGG reward surface and the PACT-family likelihood means, so this control tests posterior representation and planning under a correctly specified model by construction; it does not validate scorer accuracy under model misspecification. Prompted comparators see realized histories but do not receive the complete scorer or exhaustive planner, so cross-family ratios are system-level rather than tracker-only effects. The C1 claim is certified by the matched PACT-Joint comparison and the representation-only storage diagnostics below.

PACT-Joint agreement. The paired PACT-minus-Joint-PSRL gaps are 0.030 ± 0.294 , 0.153 ± 0.175 , -0.095 ± 0.180 , and -0.074 ± 0.346 across DeepSeek, GPT, Kimi, and Llama (mean $\pm$ SEM over common environment indices). None is resolved with ten seeds. This is consistent with, but does not prove, Proposition B.9: algorithm draws were not pathwise coupled, and provider-sampled calibration exposes no replayable sampling seed, instead pinning one realized tensor by cache.

HP-SPGG state, action, and calibration specification.

The public state at each turn records the round index and the previous turn’s realized contribution profile. Each agent’s action menu is the discrete contribution grid $\mathcal{A}_i$ with five levels, so the joint menu contains $5^3=125$ profiles that the backward-induction planner and the oracle both enumerate exactly ($\epsilon_{\text{plan}}=0$). Persona-conditioned rewards are calibrated offline once per backbone by scoring every (persona, player index, contribution profile) cell with the judge prompt of Section C.4, yielding the $125 \times 3 \times 4$ tensor that serves as both the planner’s outcome scorer and the mean of the tracker’s likelihood. The observation likelihood is the calibrated Gaussian factor with scale $\sigma=0.08$ given below Eq. (1), and the action factor is uninformative, so the online tracker issues no model calls. All reported cells use $K=20$ episodes, $n=3$ agents, $|\Theta_i|=4$ personas, and $\beta=0.25$.

Baseline definitions. *Oracle* receives the true persona profile and applies the same exhaustive joint-action argmax, providing the regret reference. Historical *PACT* samples one profile from the factored marginals, whereas the practical *PACT+* proxy plans from posterior-mean welfare plus the proxy bonus; those matched-control rows therefore differ in planner statistic as well as β . E-G supplies the clean bonus knock-out by holding the *PACT+* posterior-mean objective fixed and toggling only its bonus. *Joint-PSRL* replaces the factored persona posterior with an explicit table over all $|\Theta_i|^n$ joint profiles while sharing the likelihood, planner, and environments. *LLM-PSRL* retains a posterior-sampling loop but represents hypotheses and updates in natural language. *PSRL-NoType* samples personas without updating. *A-ToM-0/1/2* and *ECON-BNE* are our task-specific prompt adaptations of the cited methods (Mu et al. 2026; Yi et al. 2025), not executions of the original authors’ code. They receive realized histories but not the complete outcome tensor or exhaustive planner

Table 2: Environment-matched HP-SPGG control ($K=20$, 10 common seeds). Values are cumulative regret, mean $\pm$ SEM. “Best coordination” is selected post hoc by lowest mean among A-ToM-0/1/2 and ECON-BNE; its ratio to the lower-mean PACT-family row is descriptive and not a significance statistic.

Backbone	PACT ⁺	PACT	Joint-PSRL	LLM-PSRL	Best coordination	Coord./best PACT
DeepSeek-V3.2	0.360 $\pm$ 0.142	0.985 $\pm$ 0.298	0.955 $\pm$ 0.203	18.270 $\pm$ 5.141	ECON-BNE 2.550 $\pm$ 1.049	7.08 $\times$
GPT-5.4-nano	0.159 $\pm$ 0.038	0.544 $\pm$ 0.150	0.391 $\pm$ 0.100	2.193 $\pm$ 0.714	ECON-BNE 6.960 $\pm$ 1.376	43.77 $\times$
Kimi-K2.6	0.325 $\pm$ 0.088	0.547 $\pm$ 0.116	0.642 $\pm$ 0.159	6.701 $\pm$ 1.325	ECON-BNE 2.957 $\pm$ 1.366	9.10 $\times$
Llama-4-Maverick	0.430 $\pm$ 0.217	1.202 $\pm$ 0.212	1.276 $\pm$ 0.228	13.510 $\pm$ 3.431	A-ToM-0 0.700 $\pm$ 0.396	1.63 $\times$

available to the model-based PACT family. Prompt, parser, cache, and repair implementations are supplied in the code artifact; the cross-family rows should be read as system comparisons. Only PACT versus Joint-PSRL isolates posterior representation.

Representation-only comparison. Figures 9 and 10 broaden the agent-count sweep to nine baselines. Table 4 is the clean representation-only comparison: it holds the likelihood, planner, environments, and seed count fixed, and PACT and Joint-PSRL remain unresolved while the storage ratio grows from $5.3\times$ to $51\times$ over $n = 3\text{--}5$. Figure 8 is a separate $K = 20$ single-backbone diagnostic with mixed seed counts. In that figure PACT and Joint-PSRL overlap within their reported uncertainty for $n = 3, 4, 5$, but PACT⁺ fails at $n = 2$ (2.962 ± 0.545 versus 0.070 ± 0.037 for PACT). It therefore illustrates storage growth and a small-population failure of the practical bonus proxy, not uniform regret invariance across methods.

E-E: MaaSSim tracker realization. To test C1 on a deployment-shaped substrate without reusing the eight-vehicle state distribution at every scale, we use the preferred sub-fleet scheme (i): the closed-loop Nootdorp generator independently regenerates a self-consistent market for each $n \in \{2, 3, 4, 6, 8\}$ with 40 passengers, 120-second batches, and ten common environment indices. Each driver has one of 16 four-bit hidden-rule combinations. For every (n , environment) cell, factored and explicit-joint trackers consume the same saved nearest-policy accept/reject event stream in the same order, use the same product-uniform prior and analytic per-driver likelihood, and call the same exhaustive assignment objective with lexicographic tie-breaking. Their profile-sampling RNGs are intentionally independent. Conflict strength $\lambda \in \{0, 0.5, 1\}$ changes only offer travel-time/fare geometry; it does not enter the evidence stream or likelihood.

The explicit tracker is run through $n = 4$; factored tracking is run through $n = 8$. Maximum total variation between any explicit-joint marginal and its factored counterpart after any event is 2.55×10^{-14} . Eight of nine nominal paired utility CIs cover zero, including all three $\lambda = 0.5$ cells. The retained exception is $n = 2, \lambda = 0$, where joint-minus-factored utility is $+1.849$ with 95% CI $[0.192, 3.506]$. Since the posterior marginals, likelihood, event order, objective, and deterministic tie-breaking are identical, the designed source of this finite-sample difference is the independent posterior-sampling path; the cell is reported rather than averaged away. At $n = 4$, the representation is 64 versus 65,536 float64 entries, persistent belief memory is 512 bytes versus 512 KiB, and mean update time is $9.03 \mu\text{s}$ versus 0.55 ms. Factored update time remains

n	λ	Joint—factored utility [95% CI]	Max TV	Storage ratio
2	0	$+1.849 [0.192, 3.506]$	7.81×10^{-16}	$8\times$
2	.5	$-0.383 [-3.104, 2.338]$	7.81×10^{-16}	$8\times$
2	1	$+0.114 [-2.870, 3.098]$	7.81×10^{-16}	$8\times$
3	0	$+1.331 [-0.708, 3.370]$	2.47×10^{-15}	$85.3\times$
3	.5	$-3.039 [-7.237, 1.159]$	2.47×10^{-15}	$85.3\times$
3	1	$-1.589 [-5.126, 1.948]$	2.47×10^{-15}	$85.3\times$
4	0	$-1.375 [-3.901, 1.151]$	2.55×10^{-14}	$1,024\times$
4	.5	$+2.629 [-2.687, 7.945]$	2.55×10^{-14}	$1,024\times$
4	1	$-0.372 [-4.081, 3.337]$	2.55×10^{-14}	$1,024\times$

Table 3: E-E MaaSSim factored-versus-explicit-joint tracker parity (mean difference and two-sided Student- t 95% CI over ten common environment indices). Maximum TV is over every driver marginal after every event. Storage ratios are $16^n/(16n)$. The central claim is posterior identity; utility includes independent tracker sampling and has one retained nominal non-parity.

Table 4: Factored-vs-joint storage comparison on HP-SPGG ($m=|\Theta_i|=4$, $K=100$, 10 common environment seeds, $\beta=0.25$). PACT and Joint-PSRL-Uniform share likelihood, exhaustive planner, and seeds; only the posterior representation differs. Values are cumulative regret.

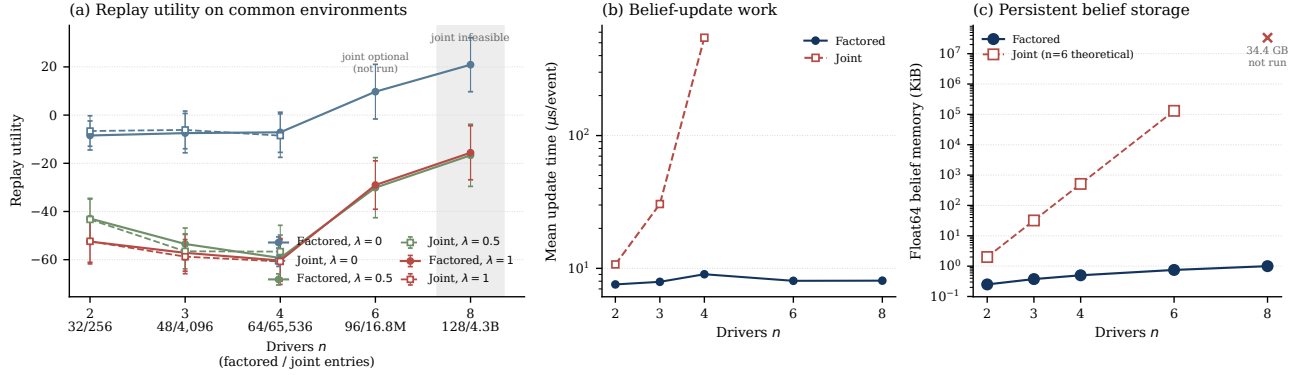
n	Storage savings	analytic kernel		DeepSeek-V3.2	
		PACT	J-PSRL-U	PACT	J-PSRL-U
3	$5.3\times$	0.12	0.17	0.57	0.38
4	$16\times$	0.05	0.11	0.80	0.51
5	$51\times$	0.60	0.63	1.02	0.85

$6.92\text{--}9.53 \mu\text{s/event}$ through $n = 8$. The optional $n = 6$ joint run is omitted; its theoretical table is 128 MiB, while $n = 8$ would require 32 GiB.

A.2 Rate and Burn-in (C2, C3)

This subsection supports C2 and evaluates C3: PACT exhibits a plateau-versus-linear separation from type-agnostic sampling on the controlled finite protocol, while the practical PACT⁺ proxy helps the matched $n = 3$ cells but is not uniformly beneficial under scaling.

HP-SPGG rate and concentration. Figure 11(a) is consistent with the rate-separation construction of Theorem B.18. Panel (b) is descriptive evidence about the practical proxy, whose formula differs from the idealized bonus in Theorem B.20. Concentration times are DeepSeek 4.4, GPT 6.6, Kimi 12.0, and Llama 14.0 episodes out of $K = 20$. The privileged analytic-kernel cross-check in Figure 12 places the proxy within 0.05 regret of the noise-free reference on every



Explicit-joint marginals match the factored tracker (max TV=2.5e-14); 8/9 nominal utility CIs include zero. Timing excludes the TV diagnostic.

Figure 7: E-E MaaSSim RQ2 diagnostic. (a) Replay utility under base, mid-conflict, and full-conflict offer geometry ($\lambda \in \{0, 0.5, 1\}$); joint and factored trackers use independent samples from numerically identical posteriors. (b) Timed Bayesian update and normalization only; the TV diagnostic and planner are excluded. (c) Persistent float64 belief-array storage. The $n = 6$ joint point is theoretical and optional; $n = 8$ is marked infeasible and was not allocated.

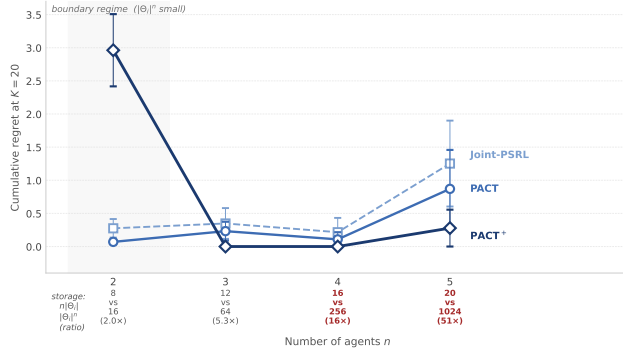


Figure 8: HP-SPGG storage-versus-regret diagnostic on the live Llama-4-Maverick tier ($|\Theta_i|=4$, $K=20$). The annotation gives $n|\Theta_i|$ factored entries versus $|\Theta_i|^n$ joint entries. PACT and Joint-PSRL have no bonus; the practical PACT+ proxy uses $\beta = 0.25$. Seed counts are mixed: $n = 2$ uses 10 seeds and $n = 3, 4, 5$ use 5. PACT and Joint-PSRL are unresolved at $n = 3, 4, 5$, whereas PACT+ has a clear $n = 2$ failure. Table 4, not this mixed-seed panel, is the representation-only comparison.

backbone in this protocol.

Iterated Concordia-derived diagnostic. The native compact Concordia evaluation is one-shot. We therefore construct a fixed-persona $K=20$ diagnostic from six configurations selected on seeds 0–4 and report on disjoint seeds 1000–1004. It retains the upstream sampler, exact payoff, and finite menu, imposes PF, and uses the same Gaussian update after reward normalization. It is Concordia-derived rather than native, and exact-payoff replay is backbone-invariant. Figure 5(a) isolates RQ2: after first averaging the six configurations within each seed, every two-sided Student- t 95% interval for PACT minus Joint-PSRL contains zero; the $K = 20$ gap is -0.017 ± 0.044 (SEM). Panel (b) isolates RQ3: PACT+ reaches 0.352 ± 0.035 cumulative and 0.0020 ± 0.0017 late regret, versus 1.361 ± 0.172 and 0.0764 ± 0.0139 for PSRL-NoType; the thin PACT reference remains near PACT+.

The two-panel trajectory appears as Figure 5 in the main

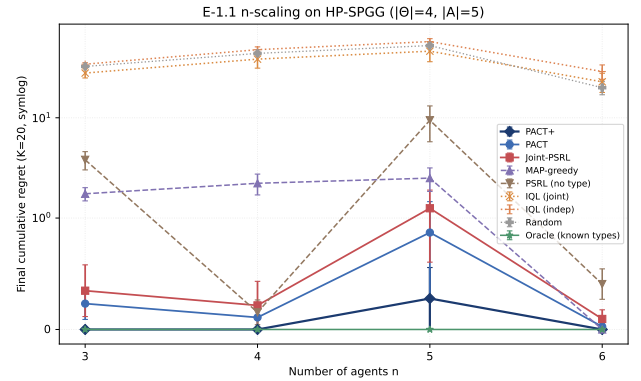


Figure 9: C1 analytic-tier nine-baseline n -scaling on HP-SPGG ($n \in \{3, 4, 5, 6\}$, $K=20$, 5 seeds, $\beta=0.25$). The PACT family stays under 1.0 cumulative regret across the sweep while type-agnostic IQL and Random grow with n . Persona-storage savings are 5.3 $\times$, 16 $\times$, 51 $\times$, and 171 $\times$ at $n = 3, 4, 5, 6$.

Table 5: Iterated Concordia-derived aggregate at $K=20$. Late is mean instant regret over episodes 11–20.

Method	Cumulative regret	Late regret
PACT	0.312 ± 0.026	0.0041 ± 0.0012
PACT+	0.352 ± 0.035	0.0020 ± 0.0017
Joint-PSRL	0.329 ± 0.030	0.0049 ± 0.0012
PSRL-NoType	1.361 ± 0.172	0.0764 ± 0.0139

paper; Table 5 records its endpoint values.

Bonus scaling and frozen coefficient. Figure 13 shows the bonus under type- and agent-space growth. Figure 14 selects $\beta = 0.25$ as the smallest value in the no-degradation interval, cutting slowest-backbone regret from 0.97 to 0.31. Table 6 holds likelihood, posterior, sampling, and policy interface fixed and changes only β on downstream substrates.

Only HP-SPGG supports interpreting the bonus as accelerated concentration. The downstream table establishes

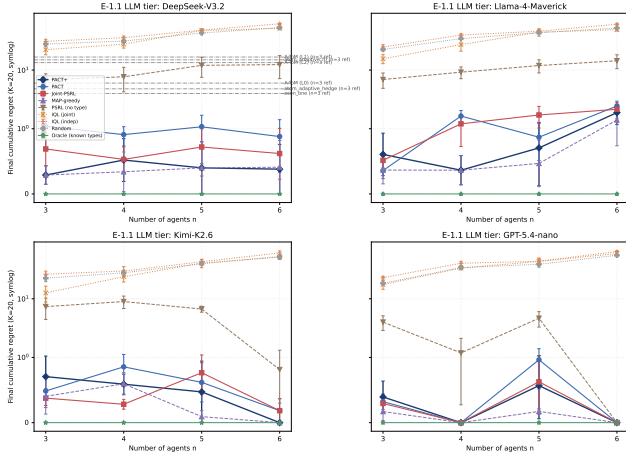


Figure 10: C1 live-LLM nine-baseline n -scaling on HP-SPGG c19 ($K=20$, 5 common environment seeds, $\beta=0.25$). Left: DeepSeek-V3.2. Right: Llama-4-Maverick. The PACT family remains separated from type-agnostic baselines on both backbones; external A-ToM-0/1/2 and ECON-BNE references are shown at $n=3$.

Table 6: C3 bonus ablations. E-G is the clean HP-SPGG knock-out: posterior-mean planner and update are fixed and only the frozen β term changes. Concordia is single-shot; historical SOTOPIA rows are descriptive because its tracker remained at the prior. MaaSSim E-F uses a separately disclosed one-step pairwise utility-disagreement bonus with $\beta = 0.25$, frozen without MaaSSim tuning.

	PACT	PACT ⁺	Δ
<i>HP-SPGG E-G (analytic, 10 environments)</i>			
Component ladder	.016 $\pm$.007	.015 $\pm$.007	−0.001
<i>Concordia (single-shot)</i>			
Pub Coordination	1.264	1.264	+0.000
Haggling (single-item)	3.661	3.661	+0.000
Haggling (multi-item)	5.429	5.429	+0.000
<i>SOTOPIA-Hard (6-turn, $n=70$/cell)</i>			
DeepSeek-V3.2	3.242	3.248	+0.006
GPT-5.4-nano	2.873	2.981	+0.107
Kimi-K2.6	2.789	2.807	+0.018
Llama-Maverick	3.308	3.308	+0.000
Mean (SOTOPIA-Hard)	3.053	3.086	+0.033
<i>MaaSSim E-F (fixed-state utility, 10 environments)</i>			
MaaSSim	27.61 $\pm$ 11.65	27.36 $\pm$ 11.56	−0.25

activity and scope: no effect in one-shot Concordia, a small descriptive SOTOPIA difference that is not update evidence, and no resolved MaaSSim gain. In E-F, PACT⁺ changes 4 of 406 assignments; paired utility difference is -0.247 ± 0.247 (SEM), with 95% CI $[-0.806, 0.312]$. Thus the frozen bonus is active but nearly inert on these offer-revealing replay states.

A.3 Structural Boundary (C4)

This subsection certifies C4: TI and RL are necessary for exact likelihood factorization, while the retained PF interventions delimit robustness without presenting a misspecified-prior

Table 7: C4 RL-violation cumulative regret at $K = 100$; Gap is the common-environment-seed PACT-minus-Joint difference.

Tier	α	PACT	Joint	Gap
Analytic	0	.001 $\pm$.001	.001 $\pm$.001	.000 $\pm$.000
Analytic	1	.155 $\pm$.065	.206 $\pm$.087	−.051 $\pm$.051
Analytic	4	.199 $\pm$.057	.213 $\pm$.057	−.013 $\pm$.013
DeepSeek live	0	1.130 $\pm$.154	1.130 $\pm$.154	.000 $\pm$.000
DeepSeek live	1	.583 $\pm$.115	.575 $\pm$.116	.008 $\pm$.008
DeepSeek live	4	.517 $\pm$.131	.466 $\pm$.108	.051 $\pm$.107

ratio as a lower-bound realization.

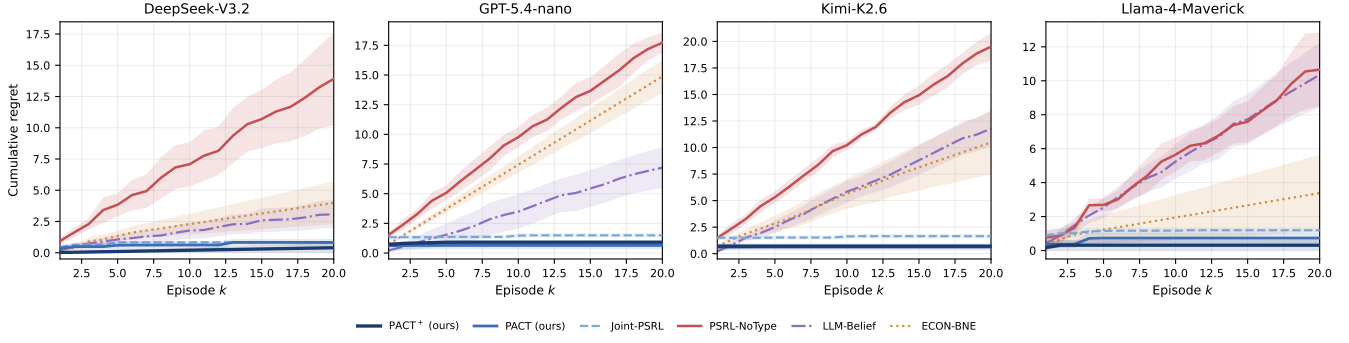
Why TI and RL are necessary. If TI fails, the transition factor depends jointly on (P, θ) and couples the transition model to the persona profile. If RL fails, for example under $r_1 = r_1^{\theta_1} - \alpha r_2^{\theta_2}$, agent 1’s likelihood contains θ_2 and couples persona coordinates at every update. Both conditions are therefore individually necessary for the factorization used by Theorem B.7.

Controlled RL-violation diagnostic. We instantiate the bounded cross-reward $r_0 = (r_0^{\theta_0} - \alpha r_1^{\theta_1} + \alpha)/(1 + \alpha)$ while other rewards remain local. PACT knows the coupled planning objective but projects its likelihood onto marginals; a shadow Joint-PSRL posterior uses the exact joint likelihood on the same trajectory. With exact enumeration, $K = 100$, and 10 common environment seeds, the methods are pathwise identical at $\alpha = 0$. As α reaches 4, marginal posterior TV rises to 0.0071 ± 0.0025 analytically and 0.0171 ± 0.0047 on a complete 324-cell live DeepSeek grid. Every paired regret gap still includes zero. Thus RL violation breaks posterior decoupling as predicted, but this geometry does not establish a monotone regret penalty.

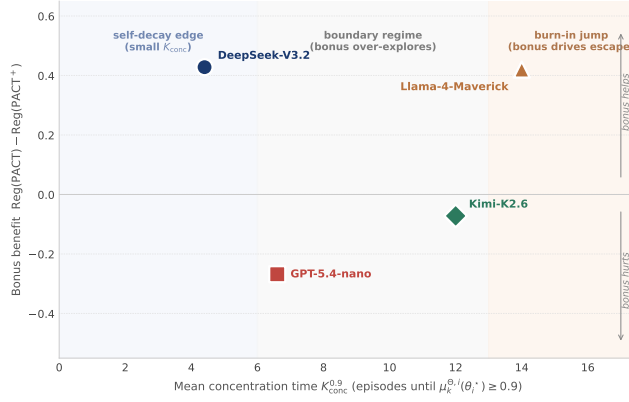
PF prior intervention. Figure 16 replaces the factored prior with correlated joint priors on the $|\Theta|^n = 64$ simplex while holding $n = 3$ and $K = 20$ fixed. A symmetric Dirichlet sweep interpolates between sparse and uniform priors; a shared-type cell is uniform over the four same-type profiles and maximally couples otherwise uniform marginals. The PACT family stays below 0.6 cumulative regret on every retained cell. Table 4 supplies the well-specified Joint-PSRL-Uniform reference; this coupled-prior result is a local finite-horizon robustness probe, not a lower-bound implementation.

True shared-type DGP. The dual intervention makes the data-generating process itself fully coupled. Joint-PSRL-Aware’s same-type prior is then well-specified while PACT’s independent marginal prior is misspecified relative to the truth.

Setup. At the start of each episode a single type $t \sim \text{Uniform}(\Theta)$ is drawn and assigned to every agent, so $\theta_i^* = t$ for all i . PACT (uniform marginals, factored representation), Joint-PSRL-Uniform (uniform Dirichlet over the $|\Theta_i|^n$ joint profiles), and Joint-PSRL-Aware (strict prior on the m same-type profiles) share the same likelihood model, backward-induction joint-action planner, and common environment seeds. We sweep $n \in \{3, 4, 5\}$ and $K \in \{50, 100\}$ on one



(a) C2: per-episode trajectory.



(b) C3: concentration speed and bonus benefit.

Figure 11: Rate and practical-proxy evidence at $n = 3$. (a) PACT and PACT⁺ plateau at < 0.05 regret per episode within $K \approx 5$ on four HP-SPGG backbones, while PSRL-NoType grows linearly and reaches an 11–30 $\times$ cumulative gap at $K=20$. (b) Vanilla-PACT concentration time $K_{\text{conc}}^{0.9}$ predicts the observed benefit of the $\beta = 0.25$ proxy in these four cells; this is not a test of the idealized bonus theorem.

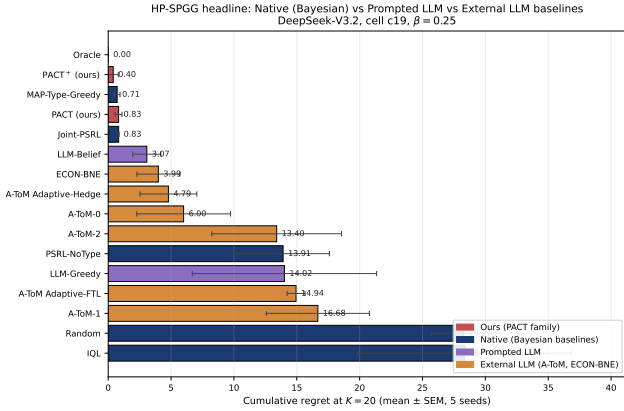


Figure 12: C2 analytic-kernel cross-check: PACT⁺ on HP-SPGG c19 remains within 0.05 regret of the privileged analytic-kernel reference across all four backbones.

analytic kernel and four live LLM backbones (DeepSeek-V3.2, GPT-5.4-nano, Kimi-K2.6, Llama-4-Maverick) with 10 common environment seeds per cell and $\beta=0.25$. Posterior storage at $n=5$ is 20 parameters for PACT, 1024 for Joint-PSRL-Uniform, and 4 for Joint-PSRL-Aware.

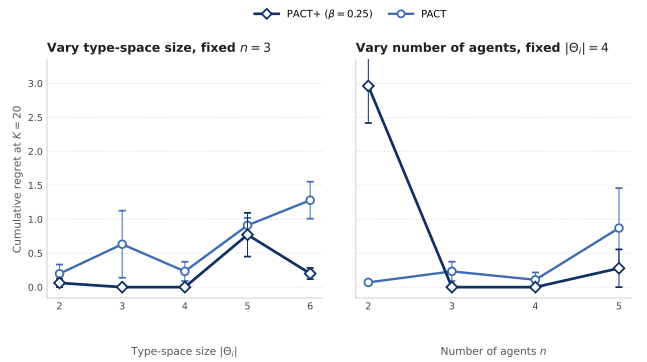


Figure 13: C3 scaling of PACT and the practical PACT⁺ proxy on HP-SPGG at $K = 20$ using the Llama-4-Maverick profile. Left varies $|\Theta_i| \in \{2, 3, 4, 5, 6\}$ at $n = 3$: type counts 2, 3, 4, 6 use 5 seeds and type count 5 uses 10. Right varies $n \in \{2, 3, 4, 5\}$ at $|\Theta_i| = 4$: $n = 2$ uses 10 seeds and $n = 3, 4, 5$ use 5. The proxy uses $\beta = 0.25$.

Reading. Under the coupled DGP, PACT remains within $1.52\times$ of the correctly supported Joint-PSRL-Aware on every reported cell and is lower-regret on 9 of 15. This is robustness under PF violation, not exact decoupling: C1’s equality claim remains restricted to TI/RL/PF, while Table 4 provides the

Table 8: True shared-type DGP robustness on HP-SPGG across one analytic kernel and four live LLM backbones ($m=|\Theta_i|=4$, $K=100$, 10 common environment seeds, $\beta=0.25$). True types are drawn from the m same-type joint profiles, so Joint-PSRL-Aware’s strict prior is well-specified and PACT’s factored prior is mis-specified relative to the coupled truth. PACT remains within $1.5\times$ of Aware on 14 of 15 cells (Kimi-K2.6 $n=4$ at $1.52\times$); the GPT-5.4-nano $n=4$ cell where Aware reaches 0 is marked with a dash. PACT is lower-regret than Aware on 9 of 15 cells.

Substrate	n	PACT	J-PSRL-A	PACT / Aware
analytic kernel	3	0.28	0.22	$1.29\times$
	4	0.00	0.32	$0.00\times$
	5	0.31	0.39	$0.79\times$
DeepSeek-V3.2	3	1.79	1.23	$1.46\times$
	4	1.31	1.84	$0.71\times$
	5	2.11	2.61	$0.81\times$
GPT-5.4-nano	3	0.64	0.74	$0.86\times$
	4	0.22	0.00	—
	5	1.64	2.36	$0.69\times$
Kimi-K2.6	3	1.14	0.86	$1.32\times$
	4	1.49	0.99	$1.52\times$
	5	1.15	1.47	$0.78\times$
Llama-4-Maverick	3	1.72	1.49	$1.16\times$
	4	1.59	1.07	$1.49\times$
	5	1.76	1.38	$1.27\times$

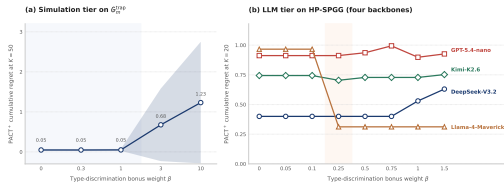


Figure 14: C3 HP-SPGG β -sweep at $K = 20$. The frozen value $\beta = 0.25$ is the smallest point in the no-degradation interval $[0.25, 0.75]$; it leaves three saturated backbones unchanged and reduces Llama-Maverick regret from 0.97 to 0.31.

representation-only comparison under the well-specified iid DGP.

A.4 Transfer Boundary (C5)

This subsection certifies C5: once actions are open-ended and TI/RL/PF cannot be verified, the exact finite-model guarantees no longer transfer, and correcting the recurrent evidence path produces no focal-score gain.

Setup recap. SOTOPIA-Hard (Zhou et al. 2024) is dyadic, open-text, judge-scored, and built from curated profile-goal pairs rather than a tested factored prior; TI/RL/PF are therefore unverified. The adapter maps utterances to a keyword-weighted intent menu and obtains proxy persona labels by applying the same mapping to full profiles. In the historical four-backbone runs, a wiring error read the inbox instead of `Observation.last_turn`, so the numeric posterior never updated; those results describe prior sampling, the surrogate, and the practical bonus proxy only. Each backbone-baseline cell contains 70 fixed hard-case episodes (14 code-names with five case instances), not 70 RNG-seeded replications, and the provider exposes no pathwise seed control. Both self-play agents use the same baseline, and focal score is the

mean of their two overall judge scores. Corrected recurrent controls are reported below.

Historical descriptive results. Table 9 reports the three private-goal families referenced in Section 4.3. Because the retained tracker did not update, the $+0.02$ – $+0.10$ margins describe the intent surrogate, prior sampling, and bonus configuration rather than recurrent Bayesian inference.

Table 9: SOTOPIA-Hard private-goal families: PACT+ vs. strongest LLM-coordination baseline per family, averaged across four backbones. Best baseline per family: A-ToM-1 for *craigslist_bargains* and *donate_funds*; ECON-BNE for *revenge_plot*.

Family	Episodes	PACT+	Best	Δ
<i>craigslist_bargains</i>	80	3.04	2.98	+0.06
<i>revenge_plot</i>	20	2.93	2.83	+0.10
<i>donate_funds</i>	20	3.26	3.24	+0.02

Aggregate scores and interpretation. Table 10 reports mean focal score by backbone. PACT+ ranks 2nd/3rd/3rd/4th on GPT/DeepSeek/Kimi/Llama and remains within 0.10 of the best alternative; all five non-oracle methods fit inside a 0.12 envelope. The one-hot *oracle_belief* row also lies within that envelope on average, whereas best-of-five *oracle_policy* reaches 3.540. Thus open action search and judge variance dominate the value of the surrogate persona belief. Because TI/RL/PF are unverified and the tracker was static, these historical rows certify a transfer boundary, not a rate or concentration claim.

Corrected recurrent tracker. We repaired the adapter to read partner evidence from SOTOPIA 0.1.5’s `Observation.last_turn`. A corrected GPT-5.4-nano run covers the same three private-goal families with 120

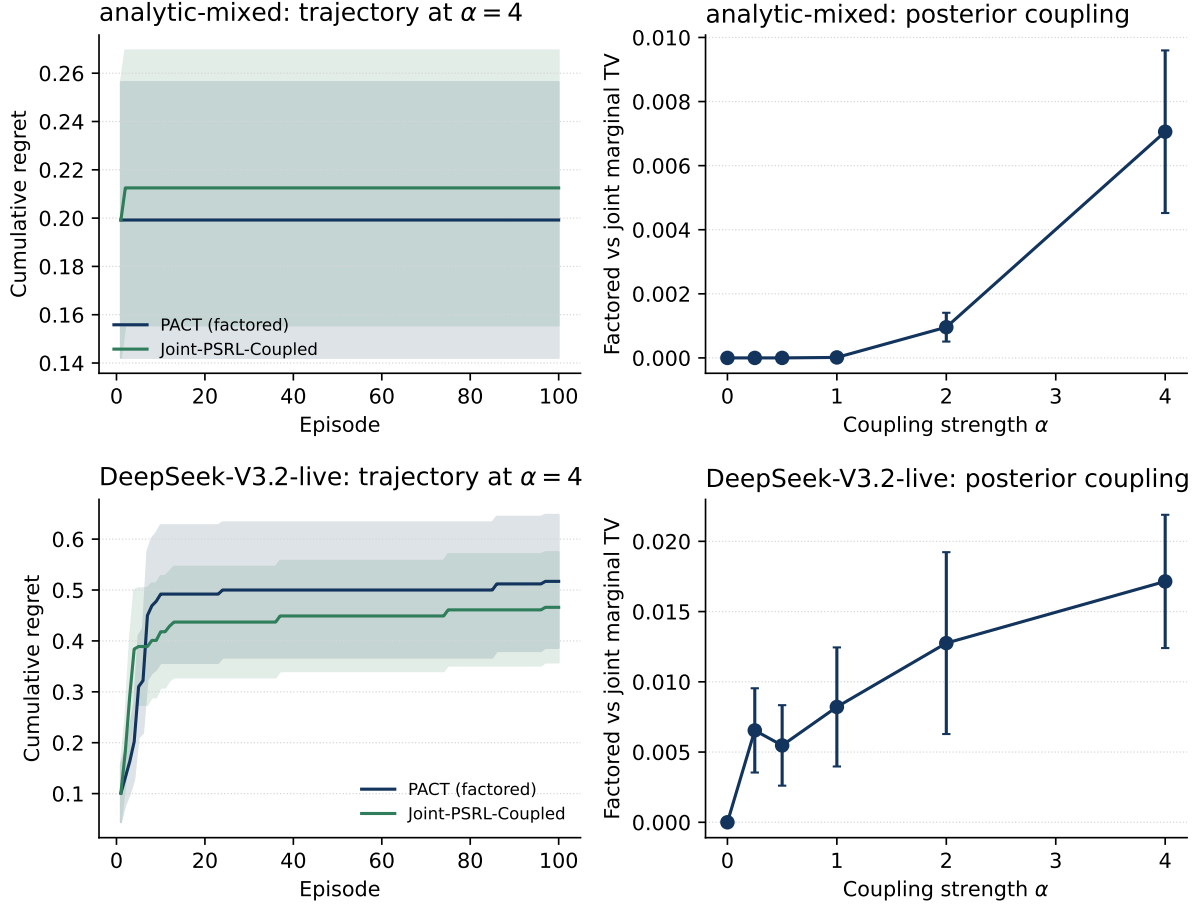


Figure 15: C4 reward-locality violation ($K=100$, 10 common environment seeds). Left: PACT and correctly specified Joint-PSRL regret trajectories at $\alpha = 4$. Right: marginal TV between PACT’s factored posterior and a shadow exact-joint posterior on the same trajectory. Top is analytic; bottom uses all 324 live DeepSeek persona/player/action cells. Posterior coupling increases, but no regret disadvantage is detectable on this geometry.

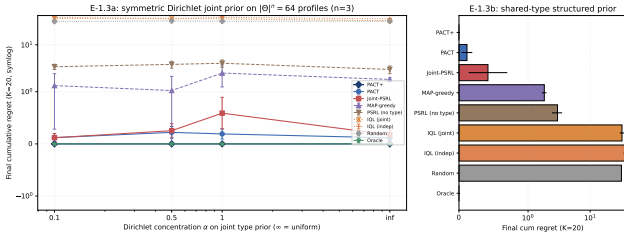


Figure 16: C4 prior swap on HP-SPGG ($n=3$, $|\Theta|=4$, $|\Theta|^n=64$ joint profiles, $K=20$, 10 common environment seeds, $\beta=0.25$, analytic calibration). Left: symmetric Dirichlet sweep on the 64-simplex; each seed draws $p \sim \text{Dir}(\alpha \mathbf{1}_{64})$ and PACT inherits the induced marginals while the explicit joint baseline uses p . Right: a shared-type prior, uniform over the four same-type profiles, leaves marginals uniform while maximally coupling them. The PACT family stays within 0.6 cumulative regret on every retained cell.

episodes at each corruption level; every accepted episode records eligible recurrent updates and zero provider-exception generation fallbacks. This run is separate from the historical four-backbone results above.

SOTOPIA provides no native labels for the project’s four persona classes. Figure 17(a) therefore evaluates concentration only against the existing *profile-derived proxy* used by the oracle adapter. Proxy mass rises from 0.25 to 0.304 ± 0.012 on *revenge_plot*, but falls to 0.233 ± 0.005 on *craigslist_bargains* and changes only to 0.261 ± 0.013 on *donate_funds*. The corrected tracker is recurrent and nontrivial, but concentration is family-dependent and is not native type-recovery accuracy.

The corrected $p=0$ focal means are 2.665 ± 0.074 , 3.229 ± 0.121 , and 2.743 ± 0.102 on *craigslist_bargains*, *donate_funds*, and *revenge_plot*. They trail the retained GPT-nano best-baseline means by 0.095, 0.157, and 0.457, respectively. We therefore select the clean-boundary branch: fixing the evidence path removes the implementation liability but does not produce a score advantage. Figure 17(b) reports the completed measured grid $p \in \{0, 0.1, 0.2, 0.25, 0.3, 0.5\}$; no point is interpolated. At $p=0.5$, changes from $p=0$ are -0.053 , $+0.161$, and $+0.146$ across the three families. The curve is non-monotone and changes are comparable to episode-row SEM, so it supports sensitivity bounds rather than a causal degradation law.

Table 10: SOTOPIA-Hard aggregate mean focal score per (backbone $\times$ baseline), 70 episodes per cell. **Bold** marks the highest non-oracle score per backbone; PACT⁺ rank and gap to best alternative are in the last two columns. The two oracle rows (Appendix A.6) receive the privileged one-hot opponent persona; `oracle_policy` additionally selects the best of $K=5$ independent episodes by the same seven-dimensional judge.

Backbone	PACT ⁺	A-ToM-1	ECON-BNE	llm_belief	llm_greedy	Rank	Δ
DeepSeek-V3.2	3.248	3.261	3.313	3.195	3.193	3	−0.065
GPT-5.4-nano	2.981	2.918	2.843	3.000	2.842	2	−0.019
Kimi-K2.6	2.807	2.733	2.876	2.906	2.761	3	−0.099
Llama-Maverick	3.308	3.359	3.319	3.340	3.165	4	−0.051
<i>Average</i>	3.086	3.068	3.088	3.110	2.990	—	−0.024

Backbone	<code>oracle_belief</code> (one-hot persona)	<code>oracle_policy</code> (one-hot + best-of- $K=5$)
DeepSeek-V3.2	3.294	3.712
GPT-5.4-nano	2.920	3.467
Kimi-K2.6	2.916	3.364
Llama-Maverick	3.282	3.618
<i>Average</i>	3.103	3.540

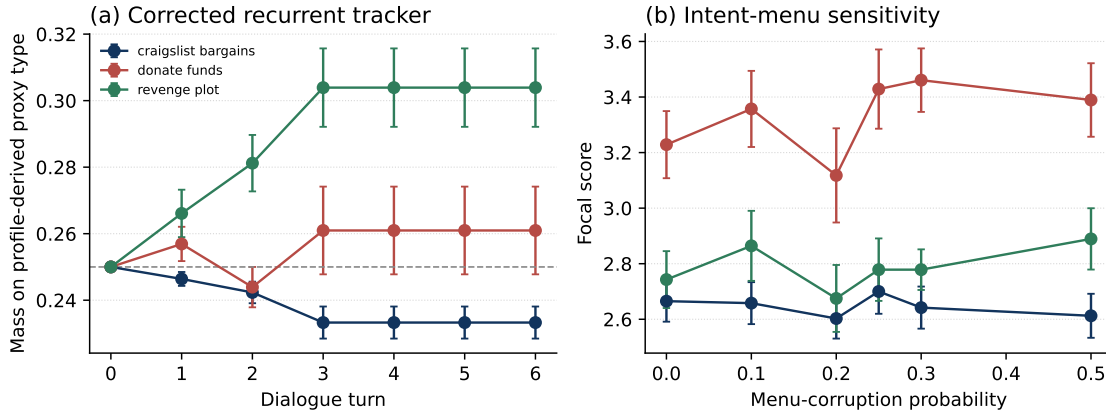


Figure 17: Corrected SOTOPIA tracker diagnostic on GPT-5.4-nano. (a) Recurrent mass on the profile-derived proxy persona (not a native SOTOPIA truth label); error bars are episode-row SEM. (b) Focal score under deterministic intent-menu corruption; provider and judge generations are not pathwise seed-controlled, so this is a sensitivity bound rather than a paired causal dose response.

Table 11: Corrected $p=0$ SOTOPIA branch decision. Historical best is the retained GPT-nano comparator, not a contemporaneous paired rerun.

Family	Corrected score	Δ vs best
Craigslist	2.665 ± 0.074	−0.095
Donate	3.229 ± 0.121	−0.157
Revenge	2.743 ± 0.102	−0.457

Corrected component controls. We rerun surrogate-only, naive-belief, and PACT⁺ on the same 30 cases and four replicate IDs. The endpoint exposes no sampling-seed control, so pairing controls case composition but not provider or judge randomness. PACT-minus-naive paired differences are $+0.044 \pm 0.096$, $+0.093 \pm 0.182$, and $−0.043 \pm 0.150$ on Craigslist, Donate, and Revenge; PACT-minus-surrogate differences are $+0.063 \pm 0.077$, $−0.079 \pm 0.090$, and $−0.107 \pm 0.166$. No lower confidence bound is positive. Together with the corrected recurrent updates, this selects the clean-boundary branch: the tracker functions, but no focal-

score gain can be attributed to its numeric update on these families.

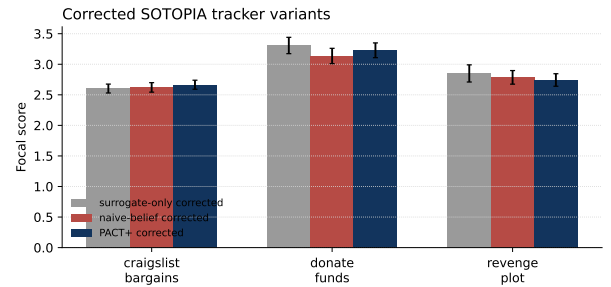


Figure 18: Corrected SOTOPIA component variants on GPT-5.4-nano (same 30 cases, four replicates; episode-row SEM). PACT⁺ does not significantly exceed either corrected control on any family. Provider generations are not pathwise seed-controlled.

Variant	Regret	Paired Δ	Ratio	95% CI for Δ
Full PACT ⁺	.015 $\pm$.007	.000	1.00 $\times$	[.000, .000]
– bonus	.016 $\pm$.007	+.001	1.05 $\times$	[–.002, .003]
– update	.675 $\pm$.114	+.660	45.61 $\times$	[.406, .915]
– identity	.700 $\pm$.368	+.685	47.29 $\times$	[–.141, 1.512]
– dispatch	6.324 $\pm$.439	+6.309	427.17 $\times$	[5.310, 7.307]

Table 12: E-G endpoint decomposition at $K = 20$ (mean $\pm$ SEM over ten common environment seeds). Δ is variant minus Full; intervals are paired two-sided Student- t 95% CIs with df 9. The bonus and identity intervals cover zero. Identity minus no-update is also unresolved: +0.025 with CI [–0.767, 0.817]. Ratios are descriptive ratios of means and are large because Full regret is near zero.

A.5 Mechanism: Posterior Identity and Centralized Dispatch (C6)

This subsection evaluates C6. E-G places the four RQ3 components on one analytic HP-SPGG substrate and metric; Concordia and MaaSSim then provide substrate-specific directional checks. In E-G, update and centralized dispatch have resolved paired effects, the identity mean moves in the expected direction but remains unresolved, and the bonus knock-out is null on this kernel.

E-G analytic component knock-out ladder. E-G uses the same deterministic analytic kernel family as Figure 9: `build_reward_tensor` with $n = 3$, backend mixed, three calibration samples, tensor seed 0, and no trap. The resulting calibrated tensor is the ground-truth reward surface and likelihood mean. All five variants share $|\Theta_i| = 4$, $K = 20$, $\sigma = 0.08$, a product-uniform prior, ten common iid-type environment seeds, and the exact centralized oracle; each variant generates its own trajectory. There are zero provider calls.

The operational knock-outs are intentionally explicit. *Full* is the practical PACT⁺ posterior-mean centralized objective at $\beta = 0.25$. *–bonus* keeps that planner and update fixed and disables only the bonus. *–update* draws a uniform profile every episode and never updates. *–identity* updates from each agent’s correct evidence, but at planning time attaches posterior rows through (2, 3, 1) on even seeds or (3, 1, 2) on odd seeds (one-based notation). *–dispatch* gives every actor an identical copy of the posterior updated from the same public action/reward history; actors independently sample profiles and independently maximize own utility, marginalizing the other simultaneous actions uniformly. Its own-agent disagreement bonus is retained, while the global joint-action-spread term is undefined because no actor selects a joint profile and is therefore omitted. This is the finite analogue of removing shared sampling and the joint argmax, not a claim that every decentralized architecture must use this action model.

E-G is the common-substrate decomposition. The remainder of this subsection supplies the older cross-substrate carriers: Concordia isolates centralized versus independent action selection, while MaaSSim probes posterior concentration and driver-identity attachment. Historical SOTOPIA-Hard is descriptive because its tracker did not update; corrected recurrent evidence is reported separately in Section A.4.

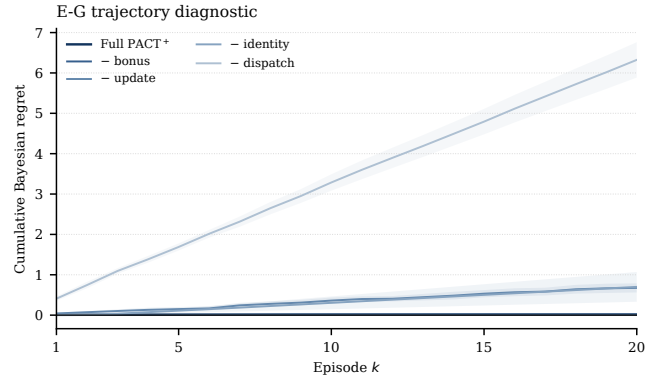


Figure 19: E-G cumulative-regret trajectories from the same 1,000-row long table as Figure 3; lines are seed means and bands are SEM. The decentralized trajectory grows almost linearly, no-update and shuffled identity remain in the same unresolved band, and Full/–bonus stay near zero.

RQ3 component	Control	Evidence
Identity attachment	fixed derangement; MaaS-Sim shuffled driver	Figure 3; Figure 22
Closed-form update	frozen uniform sampler; event decomposition	Figure 3; Figure 21
Centralized dispatch	independent best responses; prompting ablation	Figure 3; Figure 20
β bonus	same posterior-mean plan-ner, $\beta = .25$ vs. 0	Figure 3; Table 6

Table 13: RQ3 component map. E-G supplies one common-substrate control for every row; MaaSSim and Concordia are corroborating, substrate-specific carriers. Only E-G’s update and dispatch paired effects are resolved; identity is a mean-direction result and the direct bonus effect is null on this kernel.

We compare four dispatch modes sharing the same LLM on `london_mini Pub Coordination` (5 players, 2 focal, 5 seeds per backbone), two centralized (planner LLM emitting the joint action, per-agent type-guess producing n independent posteriors) and two decentralized (greedy, ToM-prompted). Within-cluster differences isolate planning-vs-belief and cluster differences isolate dispatch, with `oracle_joint` as the ceiling and full mode definitions below.

Two clusters emerge on every backbone and every metric (Figure 20). Within-cluster planning-vs-belief differences stay below 0.05 on focal payoff, while the centralized-to-decentralized gap produces a 3–4 $\times$ cumulative-regret slope difference. Joint exploration requires one planner to commit all agents to the same sampled profile; independently prompted agents miscoordinate on the states the posterior is meant to resolve. Corrected recurrent tracker controls are reported in Section A.4, so the dispatch and belief-identity mechanisms are documented separately rather than inferred from the deleted static-tracker ablation.

The remainder of this part gives the full per-backbone breakdown, the per-metric definitions, and the additional analysis.

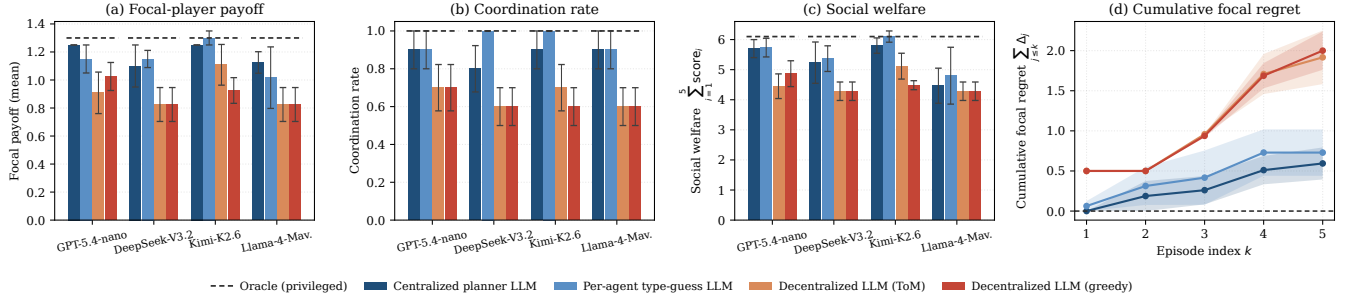


Figure 20: Price of decentralization on Concordia london_mini Pub Coordination (5 players, 2 focal, 5 scenes per cell). Centralized (planner LLM, per-agent type-guess) vs. decentralized (ToM-prompted, greedy), same LLM. (a) Focal payoff. (b) Coordination. (c) Social welfare. (d) Cumulative regret over $K=5$ scenes (bands: SE over backbones). Centralized modes cluster near `oracle_joint`; decentralized modes: -0.2 – -0.3 payoff, -0.2 – -0.4 coordination, -1 – 2 welfare units, 3 – $4\times$ cumulative regret.

Per-mode definitions. The four dispatch modes evaluated are:

- *Centralized planner LLM*, one LLM call sees the full relationship-statement matrix for all 5 players and outputs a joint posterior over every player’s favorite pub, and `argmax` yields the joint action.
- *Per-agent type-guess LLM*, n independent LLM calls, each restricted to a single player’s own social context, each outputting a posterior over that one player’s favorite pub.
- *Decentralized LLM (ToM)*, n independent LLM calls, each agent is prompted to balance own preference with theory-of-mind reasoning and pick *one* pub.
- *Decentralized LLM (greedy)*, n independent LLM calls, each agent picks the pub maximizing its own expected enjoyment with no partner inference.

Within-cluster ordering. The relative ordering inside the centralized cluster is informative for the underlying mechanism. On GPT-5.4-nano and Llama-4-Maverick the global-view planner edges the per-agent type-guess variant ($+0.10$ payoff), as expected if joint observation is the load-bearing factor. On DeepSeek-V3.2 and Kimi-K2.6 the order flips by $+0.05$. The per-agent variant slightly beats the planner because the planner occasionally splits the 2 focal players across the 2 pubs (coordination $0.80/0.90$ vs. 1.00 for the per-agent variant), while the per-agent variant defaults to the more popular pub. The structural conclusion is the same in both directions. The lift comes from the *probabilistic-reasoning prompt over a posterior on the partner’s persona*, irrespective of whether that posterior is computed once jointly or n times in parallel.

Welfare-over-all-players check. Panel (c) of Figure 20 reports the welfare sum over all 5 players (not just focal), matching the theoretical objective $W = \sum_i r_i$ used in Section 3. The centralized cluster still leads the decentralized cluster by 1 – 2 units on every backbone, so the main-text gap is not produced by NPC-side noise aligned with the focal players.

Cumulative compounding. Panel (d) shows the per-episode gap compounds. By the end of 5 episodes, decentralized modes accumulate 1.9 – 2.0 units of focal regret while

centralized modes stay at 0.6 – 0.75 , the 3 – $4\times$ slope ratio reported in the main text. This linear-in- K accumulation is the price-of-decentralization analogue of the HP-SPGG cumulative-regret divergence in Figure 2, but within a single LLM backbone. The slope difference is structural to the dispatch mode rather than to the model’s individual social-inference quality.

MaaSSim fixed-state mechanism replay. This part expands Section 4.4 only to certify C6: it isolates posterior concentration and driver-identity attachment on common saved decision states, then reports a directional single-backbone dispatch suite.

Substrate and replay protocol. Runs use the Nootdorp road-network scenario with 40 passengers, 8 vehicles, and 120-second dispatch batches. Each driver carries a four-bit hidden operational type over `{avoid_long, zone_loyal, home_pull, surge_sensitive}`, giving 16 possible rule combinations. The type is applied inside the simulator as accept/reject behavior, and the coordinator observes only which offers each driver accepts or rejects. Evaluation uses fixed-state replay. Queue snapshots and realized persona maps are saved from closed-loop runs, and every method is evaluated on the identical states, which removes the state-distribution feedback that otherwise confounds closed-loop dispatch comparisons. Realized utility combines a serve value of 3.0 per served ride, a pickup-wait cost of 0.01 per second, a driver-reject penalty of 2.0 (5.0 under stress), and a passenger-reject penalty of 0.5 .

Persona-mechanism ablation. Table 14 and Figure 22 report the mechanism ablation over 10 seeds. All PACT variants share one assignment objective and the same states, and only the driver-belief source changes, so the fixed-state comparison probes the value of the learned posterior. *PACT-prior* is MaaSSim’s substrate-specific analogue of HP-SPGG’s PSRL-NoType on the no-update axis: both freeze the uniform persona prior, although PACT-prior plans with its expectation while PSRL-NoType samples from it. The learned posterior improves realized utility over a uniform prior by $+16.50$ (27.61 vs. 11.11), closing 59.3% of the prior-to-oracle gap. The shuffled control attaches each learned posterior to the wrong driver and falls to 6.87 , below the uniform prior.

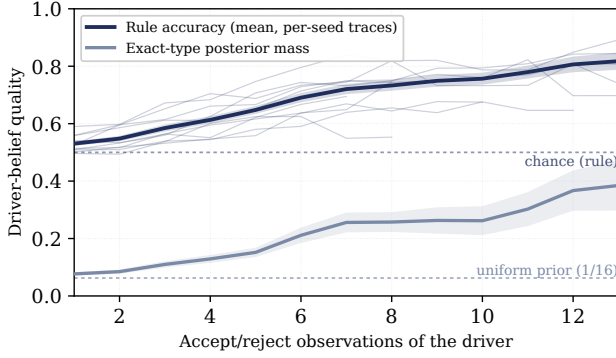


Figure 21: PACT driver-belief concentration on MaaSSim (10 seeds, thin lines are per-seed means, bands are SEM over (seed, driver) pairs). From accept/reject observations alone, marginal rule accuracy and exact-type posterior mass rise well above their chance references within a handful of observations of each driver.

Variant	Belief source	Utility	Rejects	Accept	Rule acc.
Nearest	none	11.03	26.0	0.634	—
Random	none	−33.94	30.6	0.563	—
PACT-prior	uniform	11.11	25.1	0.635	0.500
PACT-shuffled	wrong driver	6.87	26.5	0.609	0.521
PACT	learned	27.61	18.5	0.740	0.720
Oracle	true persona	38.92	13.2	0.822	1.000

Table 14: MaaSSim persona-mechanism replay (10 seeds, fixed queue snapshots and persona maps, shared assignment objective). Utility is the realized dispatch utility defined above (seed-level SEM 9.6–11.7); Rejects is mean driver rejects; Accept is the driver acceptance rate; Rule acc. is the marginal accuracy of the belief on the true driver rule.

These are directional mean differences with large seed-level SEMs, not a paired significance claim. PACT’s reported 16.25 seconds of *extra wait per snapshot* is relative to the immediate minimum-wait assignment reference; it is distinct from mean wait among served rides. Decomposing the tracker’s per-observation belief gain, rejections contribute a mean marginal rule-accuracy gain of 0.038 per event ($n=279$) against 0.022 for accepts ($n=388$). Figure 21 traces the associated concentration: marginal rule accuracy rises from chance 0.53 to 0.73 and exact-type posterior mass from 1/16 to 0.26 within eight observations, consistently across seeds.

Relative to the uniform-prior variant, PACT trades 8.1 seconds of additional mean wait among served rides (86.7 vs. 78.7) for 6.6 fewer driver rejections (18.5 vs. 25.1). The separately reported 16.25-second quantity uses the immediate minimum-wait assignment as its denominator.

LLM dispatch replay. The LLM experiments use gpt-5.4-mini (pinned snapshot 20260317) over 5 seeds with the first 20 active snapshots per seed. Each LLM policy receives a legal one-to-one assignment menu plus method-specific context and returns JSON containing an `assignment_id`; across all reported LLM methods the parse rate is 1.000 with no repairs and no fallbacks, so the action interface itself is not a confound. The prompt baselines

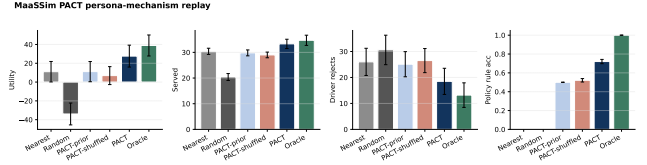


Figure 22: MaaSSim persona-mechanism replay. All variants share the same fixed queue snapshots, persona maps, and assignment objective, and only the driver-belief source changes. The learned posterior (PACT) closes 59.3% of the prior-to-oracle utility gap, while the shuffled-posterior control falls below the uniform prior, showing that correct driver-identity attachment is necessary for the gain.

Scenario	PACT	Best prompt	Gap	Oracle
Reject ($\lambda = 0$)	18.37	PSRL: 13.77	+4.60	36.44
Mid ($\lambda = .5$)	8.96	belief: −22.80	+31.75	30.70
Full ($\lambda = 1$)	8.79	belief: −30.90	+39.69	22.44

Table 15: MaaSSim LLM dispatch conflict-strength sweep (gpt-5.4-mini, 5 seeds $\times$ 20 snapshots, driver-reject penalty 5.0, parse rate 1.000). Best prompt is the strongest pure prompt baseline per level; Gap is PACT minus best prompt. Outcomes are measured reruns at $\lambda \in \{0, 0.5, 1\}$, not interpolated results.

mirror the main-text families, namely `llm_belief`, LLM-PSRL, A-ToM, and ECON-BNE. PACT is a score-assisted hybrid that receives PACT assignment scores in context, not a pure prompt method. Table 15 reports a controlled conflict-strength sweep. All cells use driver-reject penalty 5.0 and common snapshots. For each candidate offer, $\lambda \in \{0, 0.5, 1\}$ interpolates travel time and fare between the retained reject-stress offer and the full persona-risky conflict transformation; policies are rerun at every level rather than interpolating outcomes. The hybrid’s advantage over the best pure prompt baseline grows monotonically from +4.60 to +31.75 and +39.69. At $\lambda = 1$, every pure prompt baseline has negative utility (best −30.90) while the hybrid remains positive at 8.79 against an oracle at 22.44. The full-conflict endpoint is a designed stress test, not an estimate of normal operations.

Scope and limitations. Hidden types are synthetic rules rather than learned human driver preferences, and all results use one road network. The performance suite uses one LLM backbone and aggregates five seeds without retained per-seed outcomes, so no paired significance test is possible there; E-E and E-F are separate zero-provider diagnostics with retained per-environment rows. Passenger-side inference is weak because most riders are observed once (rule accuracy ≈ 0.53 against a 0.5 chance level). E-F’s bonus changes only 4/406 assignments and has an unresolved utility gap, so we claim activity and a null gain rather than a MaaSSim bonus benefit.

A.6 Oracle Constructions (Shared Infrastructure)

This subsection supports C1–C6 by certifying what each oracle reference knows and optimizes; it is shared evaluation infrastructure, not an additional empirical claim.

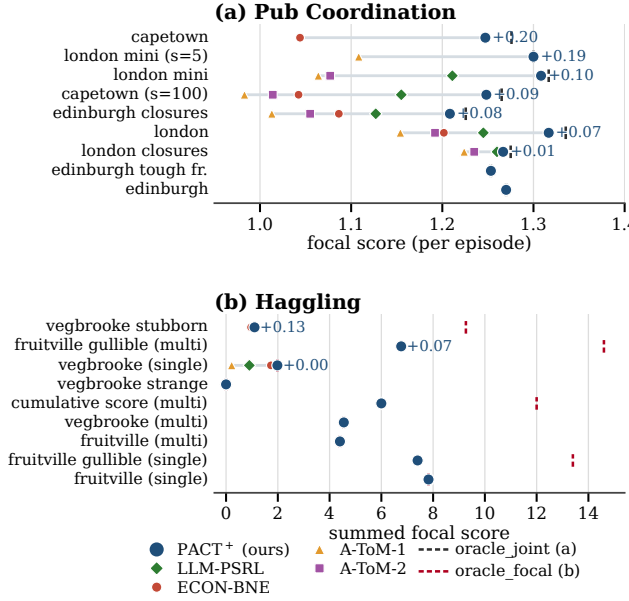


Figure 23: Full Concordia comparison across all 18 configurations. Pub Coordination uses `oracle_joint`; Hagglng uses the exact full-information focal ceiling `oracle_focal`. Rows are sorted by PACT+'s margin over the strongest baseline.

The three LLM benchmarks differ along two axes: whether payoff is closed-form and whether the action space is finite and enumerable. In every case the privileged signal is the true opponent type; the payoff or judge is part of the environment and is available to all methods.

HP-SPGG. The oracle exhaustively enumerates the finite joint contribution grid and applies a deterministic argmax to the same pinned tensor used by every method. It is therefore an exact pointwise reference for the regret calculation.

Concordia. The compact Pub and Hagglng evaluators expose a finite menu and analytic `payoff_for_case`. `oracle_joint` exactly maximizes joint welfare. Because Hagglng price is a transfer, that objective is not a focal ceiling; `oracle_focal` reruns the same exhaustive search on focal payoff and is never exceeded. Figure 24 displays the objective trade-off.

SOTOPIA-Hard. Open-ended utterances and a seven-dimensional LLM judge preclude exact enumeration. `oracle_belief` supplies the one-hot opponent persona but retains single-shot generation; it is not a strict upper bound. `oracle_policy` draws five independent episodes and retains the one with highest mean judge score. The second block of Table 10 reports both rows. Their gap measures action-search and judge-variance headroom rather than persona-information value alone.

B Theoretical Supplement

B.1 Overview and Claim-to-Theorem Map

This appendix gives the formal setting, assumptions, and proofs supporting the theoretical statements in Section 3.2–3.3.

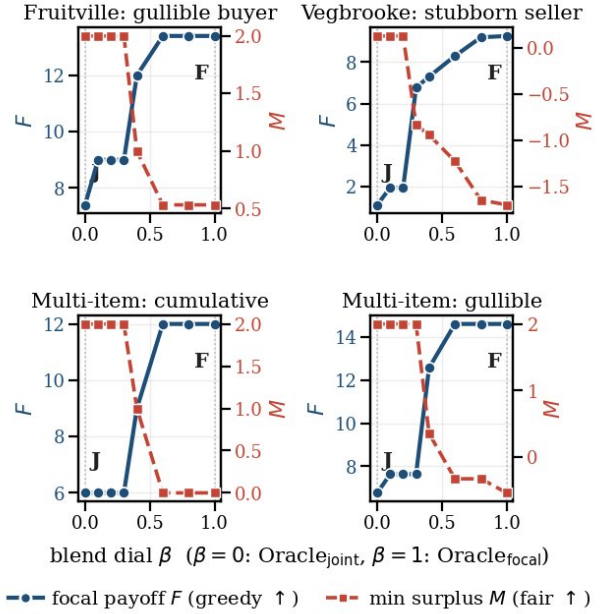


Figure 24: Focal-vs-joint Hagglng frontier. $\beta_{\text{obj}} = 0$ is `oracle_joint` (J); $\beta_{\text{obj}} = 1$ is `oracle_focal` (F). Increasing focal payoff can reduce minimum participant surplus, explaining why a focal policy may exceed the joint oracle's focal score without violating an upper bound.

Table 17 keys each main-text claim to the formal statement and proof location. The remainder is organized as: formal setting (Section B.2), posterior decoupling (Section B.3), necessity of TI and RL (Section B.4), regret bound (Section B.5), rate separation (Section B.6), idealized-bonus burn-in improvement (Section B.7), a worst-case counterexample without RL (Section B.8), and empirical validation (Section B.9).

B.2 HT-MG Formal Setting

An HT-MG is a tuple $\mathcal{G} = \langle \mathcal{S}, \{\mathcal{A}_i\}, P^*, \{\Theta_i\}, \{r_i\}, P_0, H \rangle$ with finite state space $|\mathcal{S}| = S$, joint action space $\mathcal{A} = \prod_i \mathcal{A}_i$ of size A_{joint} , type space $\Theta = \prod_i \Theta_i$, transition kernel $P^*: \mathcal{S} \times \mathcal{A} \times \Theta \rightarrow \Delta(\mathcal{S})$, per-agent reward $r_i: \mathcal{S} \times \mathcal{A} \times \Theta_i \rightarrow [0, 1]$, joint prior P_0 , and horizon H . At each episode a type profile $\theta^* \in \Theta$ is sampled once and held fixed. P^* is fixed but unknown. A centralized learner observes the trajectory τ_k and updates a posterior over (P^*, θ^*) .

Assumption B.1 (TI). $P^*(s' | s, a, \theta) = P^*(s' | s, a)$.

Assumption B.2 (RL). $r_i(s, a, \theta) = r_i(s, a, \theta_i)$.

Assumption B.3 (PF). $P_0(P, \theta) = P_0^P(P) \prod_i P_0^{\Theta, i}(\theta_i)$.

Assumption B.4 (TID). For the per-agent outcome channel $q_{\theta_i}(\cdot | s, a)$, $\inf_{s, a} D_H^2(q_{\theta_i}(\cdot | s, a), q_{\theta'_i}(\cdot | s, a)) \geq \rho > 0$ for all $\theta_i \neq \theta'_i$.

Assumption B.5 (ISI). Each s_1^k is drawn from a fixed distribution ν_0 independently of (P^*, θ^*) and of the episode- k sampling randomness.

Assumption B.6 (DP). $\mathcal{P}$ is a measurable deterministic mapping from a model to a centralized joint policy.

Table 16: Oracle construction across the three LLM benchmarks.

Benchmark	Payoff form	Action space	Oracle type
HP-SPGG	analytic/pinned tensor	finite, 125 joint profiles	strict exhaustive argmax
Concordia	analytic, closed-form	finite menu	strict joint/focal argmax
SOTOPIA-Hard	7-dimensional LLM judge	open-ended utterance	non-strict belief reference + best-of- K heuristic

Table 17: Claim-to-theorem map. Each main-text statement is keyed to the formal theorem and the appendix subsection containing the proof.

Main-text claim	Formal statement	Proof location
Exact factorization and persona-belief complexity	Thm. B.7, Cor. B.8	§B.3
TI and RL are both necessary	Factorization argument + controlled diagnostic	§A.3; theory pointer §B.4
$\tilde{O}(\sqrt{K})$ regret bound for PACT	Thm. B.12	§B.5
Rate separation vs PSRL-NoType	Thm. B.18	§B.6
Idealized-bonus burn-in improvement	Thm. B.20	§B.7
Worst-case burn-in without RL	Thm. B.21	§B.8

The type-conditional Bayesian regret of agent i over K episodes is

$$\text{Reg}_i^B(K) := \mathbb{E}_{(P^*, \theta^*) \sim P_0} \sum_{k=1}^K \left(V_i^{\mathcal{P}(P^*, \theta^*)}(s_1^k; \theta_i^*) - V_i^{\pi_k}(s_1^k; \theta_i^*) \right). \quad (4)$$

B.3 Posterior Decoupling

Theorem B.7 (Posterior Decoupling). *Under Assumptions B.1, B.2, B.3, the posterior factorizes at every episode:*

$$P(P, \theta \mid \mathcal{H}_k) = \mu_k^P(P) \cdot \prod_{i \in [n]} \mu_k^{\Theta, i}(\theta_i), \quad (5)$$

where $\mathcal{H}_k$ is the history before episode k ; observing episode k produces the update to μ_{k+1} .

Proof. Conditional on π_k , the trajectory likelihood is a product of per-step transition, action, and reward factors. TI drops θ from the transition factor. RL localizes each action and reward factor to θ_i , giving $L_{\Theta, i}(\tau_k; \theta_i) = \prod_h \ell_{i, h}^a(\theta_i) P(r_h^{k, i} \mid s_h^k, a_h, \theta_i)$. Equation (1) applies Bayes’ rule separately to these local factors, and induction from PF yields the stated posterior. In HP-SPGG the action factor is uninformative and the reward factor is Gaussian. Open-text SOTOPIA-Hard does not satisfy the finite calibrated-outcome assumptions; its intent-space surrogate is an empirical adapter and is not covered by this theorem. $\square$

Corollary B.8 (Persona-posterior complexity). *Excluding the transition-posterior term common to both representations, PACT requires $O(\sum_i |\Theta_i|)$ persona-posterior storage and candidate-likelihood evaluations per update, whereas an explicit joint table requires $O(\prod_i |\Theta_i|)$. For uniform persona libraries this is $O(n|\Theta_i|)$ versus $O(|\Theta_i|^n)$.*

Proposition B.9 (Pathwise equivalence with Joint-PSRL). *Under matched RNG and the assumptions of Theorem B.7, PACT and Joint-PSRL produce pathwise-identical trajectories almost surely. In particular $\text{Reg}_i^B[\text{PACT}](K) = \text{Reg}_i^B[\text{Joint-PSRL}](K)$ for every i, K .*

B.4 Necessity of (TI) and (RL)

The necessity argument and controlled RL-violation diagnostic have been moved to the claim-organized structural-boundary evidence in Section A.3; this label is retained as the theory-side pointer.

B.5 Regret Bound

This subsection proves the regret bound of PACT. The argument has three components. First, posterior decoupling (Theorem B.7) implies that sampling the transition posterior and the n persona marginals independently constitutes an exact draw from the joint posterior, so PACT inherits the posterior-matching property of posterior-sampling RL while storing only $O(n|\Theta_i|)$ persona entries (Lemma B.10). Second, the per-episode regret decomposes into a planner-approximation term, a transition channel, and a per-agent reward channel (Lemma B.13). Third, the two channels are bounded separately. The transition channel is bounded by Dirichlet posterior concentration (Lemma B.14) and the reward channel by a self-contained information-ratio argument on the finite type space Θ_i (Lemma B.15), which is where the per-agent rate $O(H\sqrt{|\Theta_i| \mathcal{H}(P_0^{\Theta, i}) K})$ is obtained. The rate depends on the library size $|\Theta_i|$ and the entropy $\mathcal{H}(P_0^{\Theta, i})$ of agent i ’s persona prior rather than on the joint space $|\Theta_i|^n$, so an informative prior provably reduces regret.

Notation. For a kernel P , a policy π , and a reward parameter $\theta_i \in \Theta_i$, the stage value of agent i is $V_{i, h}^{\pi_h}(s; P, \theta_i) := \mathbb{E}^{P, \pi}[\sum_{h'=h}^H r_i(s_{h'}, a_{h'}, \theta_i) \mid s_h = s]$ and $V_i^\pi := V_{i, 1}^\pi$. In Eq. (4), $V_i^\pi(s; \theta_i^*)$ abbreviates $V_i^\pi(s; P^*, \theta_i^*)$. Write $M^* := (P^*, \theta^*)$ for the true world, $\hat{M}_k := (\hat{P}_k, \hat{\theta}_k)$ for the episode- k sample of Algorithm 1 with $\beta = 0$, $\mathcal{H}_k$ for the history before episode k , and $\mathbb{E}_k[\cdot] := \mathbb{E}[\cdot \mid \mathcal{H}_k]$. Let $N_k(s, a)$ count the visits to (s, a) before episode k and let $\bar{P}_k(\cdot \mid s, a) := \mathbb{E}_k[P^*(\cdot \mid s, a)]$ denote the posterior mean kernel. Let $\hat{\mathcal{P}}$ denote the implemented planner and assume $V_i^{\hat{\mathcal{P}}(M)}(s; M) \geq V_i^{\mathcal{P}(M)}(s; M) - \epsilon_{\text{plan}}$ relative to the fixed exact mapping in (DP). The finite HP-SPGG and compact Concordia planners enumerate their menus exactly, hence $\epsilon_{\text{plan}} = 0$; SOTOPIA-Hard lies outside this finite-planner theorem. Since (TI) makes the transition persona-blind, the only type-dependent observation is the per-agent per-turn outcome channel (emitted action and realized reward), and

Assumption B.4 is read on that channel, written $q_{\theta_i}(\cdot \mid s, a)$ below.

Lemma B.10 (Factored sampling is exact posterior sampling). *Under Assumption 3.1, the pair $\hat{M}_k$ produced by drawing $\hat{P}_k \sim \mu_k^P$ and $\hat{\theta}_i^k \sim \mu_k^{\Theta, i}$ independently across i satisfies $\hat{M}_k \sim P(\cdot \mid \mathcal{H}_k)$. Conditionally on $\mathcal{H}_k$, the sample $\hat{M}_k$ and the true world M^* are independent and identically distributed.*

Proof. By Theorem B.7 the joint posterior of (P^*, θ^*) given $\mathcal{H}_k$ is the product measure $\mu_k^P \otimes \bigotimes_i \mu_k^{\Theta, i}$. Algorithm 1 draws the coordinates independently from exactly these marginals, so the conditional law of $\hat{M}_k$ given $\mathcal{H}_k$ coincides with the posterior. Independence from M^* holds because the sampling randomness is exogenous. $\square$

Lemma B.11 (Posterior matching). *For every bounded measurable g and every k , $\mathbb{E}_k[g(M^*)] = \mathbb{E}_k[g(\hat{M}_k)]$. Under (ISI) the identity also holds conditionally on $(\mathcal{H}_k, s_1^k)$.*

Proof. Immediate from Lemma B.10. Under (ISI) the initial state s_1^k is independent of M^* and of the sampling randomness, so conditioning on it changes neither the posterior nor the sampling law. $\square$

Theorem B.12 (Bayesian Regret of PACT). *Under Assumptions B.1, B.2, B.3, B.5, and B.6, and a Dirichlet prior on P^* ,*

$$\mathbb{E}[\text{Reg}_i^B(K)] \leq \tilde{O}(H^{3/2} \sqrt{SA_{\text{joint}} K}) + O(H \sqrt{|\Theta_i| \mathcal{H}(P_0^{\Theta, i}) \cdot K}) + K \epsilon_{\text{plan}}, \quad (6)$$

where $\mathcal{H}(P_0^{\Theta, i})$ is the Shannon entropy of agent i 's persona prior.

Since $\mathcal{H}(P_0^{\Theta, i}) \leq \log |\Theta_i|$ with equality iff the prior is uniform, the uniform-prior form $O(H \sqrt{|\Theta_i| \log |\Theta_i| K})$ is the worst case of the persona term. The prior enters the bound only through its entropy, so any prior information about agent i 's persona translates into a provable regret reduction.

Lemma B.13 (Regret decomposition). *Under (RL), (ISI), and (DP) with an ϵ_{plan} -approximate planner,*

$$\text{Reg}_i^B(K) \leq K \epsilon_{\text{plan}} + \mathbb{E} \sum_{k=1}^K (\Delta_k^P + \Delta_k^\Theta),$$

where, writing $\hat{\theta}_i^k$ for the i -th coordinate of $\hat{\theta}_k$,

$$\begin{aligned} \Delta_k^P &:= V_i^{\pi_k}(s_1^k; \hat{P}_k, \hat{\theta}_i^k) - V_i^{\pi_k}(s_1^k; P^*, \hat{\theta}_i^k), \\ \Delta_k^\Theta &:= V_i^{\pi_k}(s_1^k; P^*, \hat{\theta}_i^k) - V_i^{\pi_k}(s_1^k; P^*, \theta_i^*). \end{aligned}$$

Proof. Write $\pi_k^* := \mathcal{P}(M^*)$ for the comparator in Eq. (4) and add and subtract $V_i^{\pi_k}(s_1^k; \hat{M}_k)$, so that the per-episode regret equals

$$(V_i^{\pi_k}(s_1^k; M^*) - V_i^{\pi_k}(s_1^k; \hat{M}_k)) + (V_i^{\pi_k}(s_1^k; \hat{M}_k) - V_i^{\pi_k}(s_1^k; M^*)).$$

For the first bracket, define $g(M) := V_i^{\mathcal{P}(M)}(s_1^k; M)$ with $\mathcal{P}$ the exact deterministic mapping of (DP). The map g is measurable and bounded by H . By Lemma B.11, $\mathbb{E}[g(M^*) \mid \mathcal{H}_k, s_1^k] = \mathbb{E}[g(\hat{M}_k) \mid \mathcal{H}_k, s_1^k]$, and by the planner convention

$V_i^{\pi_k}(s_1^k; \hat{M}_k) \geq g(\hat{M}_k) - \epsilon_{\text{plan}}$. Taking expectations, the first bracket contributes at most ϵ_{plan} per episode. The second bracket splits into $\Delta_k^P + \Delta_k^\Theta$ by adding and subtracting $V_i^{\pi_k}(s_1^k; P^*, \hat{\theta}_i^k)$, where (RL) guarantees that agent i 's value depends on $\hat{\theta}_i^k$ only through $\hat{\theta}_i^k$. $\square$

Lemma B.14 (Transition channel). *Under a Dirichlet prior on P^* , $\mathbb{E} \sum_{k=1}^K \Delta_k^P \leq \tilde{O}(H^{3/2} \sqrt{SA_{\text{joint}} K})$.*

Proof. Fix k and abbreviate $\hat{V}_{h+1}(\cdot) := V_{i, h+1}^{\pi_k}(\cdot; \hat{P}_k, \hat{\theta}_i^k) \in [0, H - h]$. The standard simulation identity (Osband, Russo, and Van Roy 2013) gives

$$\Delta_k^P = \sum_{h=1}^H \mathbb{E}^{P^*, \pi_k} [\langle \hat{P}_k(\cdot \mid s_h, a_h) - P^*(\cdot \mid s_h, a_h), \hat{V}_{h+1} \rangle],$$

where the outer expectation is over the trajectory executed under the true kernel. Let $\beta_k(s, a) := \sqrt{14S \log(2SA_{\text{joint}} HK) / \max\{1, N_k(s, a)\}}$ and let B_k be the set of kernels P with $\|P(\cdot \mid s, a) - \bar{P}_k(\cdot \mid s, a)\|_1 \leq \beta_k(s, a)$ for all (s, a) . By the ℓ_1 concentration inequality of Weissman et al. (2003) applied to the transition samples collected at each (s, a) , combined with the stopping-time argument of Auer, Jaksch, and Ortner (2008) to handle adaptive data collection, $\Pr(P^* \notin B_k) \leq 1/(KH)$, where the $O(S/\max\{1, N_k\})$ smoothing gap between the Dirichlet posterior mean and the empirical mean is absorbed into the constant. Since B_k is $\mathcal{H}_k$ -measurable, Lemma B.11 gives $\Pr(\hat{P}_k \notin B_k) = \Pr(P^* \notin B_k)$. On the event $\{P^* \in B_k, \hat{P}_k \in B_k\}$, centring $\hat{V}_{h+1}$ at its midpoint yields, uniformly over state-action pairs,

$$|\langle \hat{P}_k - P^*, \hat{V}_{h+1} \rangle(s_h, a_h)| \leq H \beta_k(s_h, a_h),$$

and the uniformity over (s, a) makes the bound valid despite the adaptivity of the visitation sequence. Off the event, $|\Delta_k^P| \leq H$, which contributes at most $2H$ over all K episodes at the chosen failure probability. The pigeonhole bound over visitation counts, with counts frozen within an episode (Auer, Jaksch, and Ortner 2008; Osband, Russo, and Van Roy 2013), gives $\sum_{k, h} \max\{1, N_k(s_h^k, a_h^k)\}^{-1/2} \leq (\sqrt{2} + 1) \sqrt{SA_{\text{joint}} KH} + SA_{\text{joint}} H$. Combining,

$$\mathbb{E} \sum_k \Delta_k^P \leq \tilde{O}(H^{3/2} S \sqrt{A_{\text{joint}} K}).$$

This elementary route carries an extra $\sqrt{S}$ factor relative to the stated rate. The stated rate follows by replacing the ℓ_1 step with the Gaussian–Dirichlet stochastic-optimism analysis of Osband and Van Roy (2017), which bounds the inner product $\langle \hat{P}_k - P^*, \hat{V}_{h+1} \rangle$ directly at $\tilde{O}(H/\sqrt{\max\{1, N_k(s, a)\}})$ per visited pair, without paying the ℓ_1 union over next states. That analysis requires the sample to come from the exact posterior under a Dirichlet prior, which Lemma B.10 supplies. Repeating the pigeonhole step then yields $\mathbb{E} \sum_k \Delta_k^P \leq \tilde{O}(H^{3/2} \sqrt{SA_{\text{joint}} K})$. $\square$

Lemma B.15 (Reward channel). *Under Assumption 3.1, (ISI), and (DP),*

$$\mathbb{E} \sum_{k=1}^K \Delta_k^\Theta \leq H \sqrt{\frac{1}{2} |\Theta_i| \mathcal{H}(P_0^{\Theta,i}) K}.$$

Proof. Condition on $\mathcal{H}_k$ and write $\nu := \mu_k^{\Theta,i}$, which is both the posterior of θ_i^* and the sampling law of $\hat{\theta}_i^k$ by Lemma B.10. Let $\xi_k := (\hat{P}_k, \hat{\theta}_{-i}^k, s_1^k)$ collect the remaining episode- k randomness. By Theorem B.7, (ISI), and independent sampling, the variables ξ_k , $\hat{\theta}_i^k$, θ_i^* , and P^* are mutually independent given $\mathcal{H}_k$. For $\theta, \theta' \in \Theta_i$ define

$$G(\theta, \theta') := \mathbb{E}_k \left[\sum_{h=1}^H r_i(s_h^k, a_h^k, \theta') \mid \hat{\theta}_i^k = \theta \right] \in [0, H],$$

where the trajectory is generated by executing $\pi_k = \mathcal{P}(\hat{P}_k, (\theta, \hat{\theta}_{-i}^k))$ under P^* and the expectation integrates over ξ_k and P^* . This quantity is well defined because, under centralized dispatch and (TI), the trajectory law depends on personas only through the sampled profile entering the planner and not through θ^* . Since $\hat{\theta}_i^k \perp \theta_i^* \mid \mathcal{H}_k$ with common law ν ,

$$\mathbb{E}_k[\Delta_k^\Theta] = \sum_{\theta} \nu(\theta) G(\theta, \theta) - \sum_{\theta, \theta'} \nu(\theta) \nu(\theta') G(\theta, \theta') = \sum_{\theta} \nu(\theta) (G(\theta, \theta) - \bar{G}(\theta)),$$

where $\bar{G}(\theta) := \sum_{\theta'} \nu(\theta') G(\theta, \theta')$. Let $R_k := \sum_h r_{i,h}^k \in [0, H]$ denote the realized episode reward of agent i , an observed quantity, and let $I_k := I(\theta_i^*; (\hat{\theta}_i^k, R_k))$ with mutual information computed under the conditional law given $\mathcal{H}_k$. Since $\hat{\theta}_i^k \perp \theta_i^* \mid \mathcal{H}_k$, the chain rule gives $I_k = I(\theta_i^*; R_k \mid \hat{\theta}_i^k)$.

Step 1 (Cauchy–Schwarz). By Cauchy–Schwarz over Θ_i ,

$$(\mathbb{E}_k[\Delta_k^\Theta])^2 \leq |\Theta_i| \sum_{\theta} \nu(\theta)^2 (G(\theta, \theta) - \bar{G}(\theta))^2.$$

Step 2 (Pinsker). Given $\hat{\theta}_i^k = \theta$, the conditional law of R_k given additionally $\theta_i^* = \theta'$ has mean $G(\theta, \theta')$ and support in $[0, H]$, while its θ' -marginal mixture has mean $\bar{G}(\theta)$. Pinsker’s inequality combined with $|\mathbb{E}X - \mathbb{E}Y| \leq H \cdot \text{TV}$ for $[0, H]$ -valued variables gives

$$\text{KL}(\mathcal{L}(R_k \mid \hat{\theta}_i^k = \theta, \theta_i^* = \theta') \parallel \mathcal{L}(R_k \mid \hat{\theta}_i^k = \theta)) \geq \frac{2}{H^2} (G(\theta, \theta') - \bar{G}(\theta))^2.$$

Averaging over $\theta \sim \nu$ and $\theta' \sim \nu$ and keeping only the diagonal terms $\theta' = \theta$,

$$I_k \geq \frac{2}{H^2} \sum_{\theta, \theta'} \nu(\theta) \nu(\theta') (G(\theta, \theta') - \bar{G}(\theta))^2 \geq \frac{2}{H^2} \sum_{\theta} \nu(\theta)^2 (G(\theta, \theta) - \bar{G}(\theta))^2.$$

Combining Steps 1 and 2 yields the information-ratio bound $(\mathbb{E}_k[\Delta_k^\Theta])^2 \leq \frac{H^2 |\Theta_i|}{2} I_k$, the finite-class bound of Russo and Van Roy (2016) instantiated on the type space with the truth θ_i^* in the role of the target variable. The argument uses only the algebraic structure above, so optimality of the target is not required.

Step 3 (information budget). The pair $(\hat{\theta}_i^k, R_k)$ is measurable with respect to the episode- k increment of the history augmented with the coordinator’s internal sampling randomness, which is independent of θ_i^* . By the data-processing inequality and the chain rule for mutual information,

$$\sum_{k=1}^K \mathbb{E}[I_k] \leq I(\theta_i^*; \mathcal{H}_{K+1}) \leq \mathcal{H}(\theta_i^*) = \mathcal{H}(P_0^{\Theta,i}),$$

where the final equality holds because θ_i^* is drawn from $P_0^{\Theta,i}$. By Cauchy–Schwarz over episodes,

$$\mathbb{E} \sum_k \Delta_k^\Theta \leq \sqrt{K \cdot \mathbb{E} \sum_k (\mathbb{E}_k[\Delta_k^\Theta])^2} \leq \sqrt{K \cdot \frac{H^2 |\Theta_i|}{2} \mathcal{H}(P_0^{\Theta,i})}.$$

The resulting bound depends only on the per-agent library size $|\Theta_i|$ and the entropy of the per-agent prior, not on the joint space $\prod_j |\Theta_j|$. Decoupling enters twice, once to license the marginal sampler (Lemma B.10) and once to remove every other agent’s type from the conditional analysis through the independence of ξ_k . $\square$

Proof of Theorem B.12. Combine Lemma B.13 with Lemmas B.14 and B.15. $\square$

Proposition B.16 (Geometric collapse of the reward channel under TID). *Under the assumptions of Theorem B.12, Assumption B.4 with gap $\rho > 0$ on the per-turn outcome channel, and a prior with $\pi_{\min} := \min_{\theta} P_0^{\Theta,i}(\theta) > 0$,*

$$\mathbb{E} \sum_{k=1}^K \Delta_k^\Theta \leq O\left(H + \rho^{-1} \log \frac{|\Theta_i|}{\pi_{\min}}\right),$$

a constant independent of K . Transition learning is then the asymptotically dominant term of Eq. (6) at rate $\tilde{O}(\sqrt{K})$.

Proof. For a wrong type $\theta' \neq \theta_i^*$ let $\Lambda_k(\theta') := \mu_k^{\Theta,i}(\theta') / \mu_k^{\Theta,i}(\theta_i^*)$ denote the posterior odds. Bayes’ rule gives $\Lambda_{k+1}(\theta') = \Lambda_k(\theta') \prod_{h=1}^H q_{\theta'}(y_h \mid s_h, a_h) / q_{\theta_i^*}(y_h \mid s_h, a_h)$ with y_h the turn- h outcome of agent i . Conditionally on (s_h, a_h) and the past, the Hellinger affinity of the per-turn likelihood ratio satisfies

$$\mathbb{E} \left[\sqrt{q_{\theta'}(y_h \mid s_h, a_h) / q_{\theta_i^*}(y_h \mid s_h, a_h)} \right] = 1 - D_H^2(q_{\theta'}, q_{\theta_i^*}) \leq e^{-\rho}$$

by Assumption B.4, uniformly over (s, a) , and the uniformity handles the adaptive visitation (Ghosal and Van der Vaart 2017). Hence $\sqrt{\Lambda_k(\theta')}$ is a nonnegative supermartingale contracting by $e^{-\rho}$ per turn, so $\mathbb{E} \sqrt{\Lambda_k(\theta')} \leq \pi_{\min}^{-1/2} e^{-\rho H(k-1)}$. Since $\mu_k(\theta') \leq \min\{1, \Lambda_k(\theta')\} \leq \sqrt{\Lambda_k(\theta')}$, this gives $\mathbb{E} \mu_k^{\Theta,i}(\theta') \leq \pi_{\min}^{-1/2} e^{-\rho H(k-1)}$.

The reward channel vanishes on the event $\{\hat{\theta}_i^k = \theta_i^*\}$ because Δ_k^Θ compares the same policy under the same kernel with identical reward parameters, and $|\Delta_k^\Theta| \leq H$ otherwise. Averaging the truth over the posterior, $\Pr_k(\hat{\theta}_i^k \neq \theta_i^*) = \mathbb{E}_k[\sum_{\theta' \neq \theta_i^*} \mu_k^{\Theta,i}(\theta')]$, so

$$\mathbb{E}[\Delta_k^\Theta] \leq H \min\{1, (|\Theta_i| - 1) \pi_{\min}^{-1/2} e^{-\rho H(k-1)}\}.$$

Let $C := |\Theta_i| \pi_{\min}^{-1/2}$ and $k_0 := \lceil 1 + \log C / (\rho H) \rceil$. The first k_0 episodes contribute at most $H k_0 = O(H + \rho^{-1} \log C)$ and the geometric tail contributes at most $H / (1 - e^{-\rho H}) \leq H + \rho^{-1}$. Summing proves the claim. For the uniform prior, $\log C = O(\log |\Theta_i|)$, recovering the $O(H \log |\Theta_i| / \rho)$ collapse cited in the main text as a weaker corollary. $\square$

Remark B.17 (Prior information is double-edged). *The persona prior enters the two regimes of the reward channel in opposite directions. In the $\sqrt{K}$ regime of Lemma B.15 the channel scales with $\sqrt{\mathcal{H}(P_0^{\Theta,i})}$, so a more informative prior provably reduces regret. In the TID regime of Proposition B.16 the burn-in constant scales with $\log(1/\pi_{\min})$, so a prior that is sharp in the wrong direction, placing near-zero mass on the realized persona, inflates the constant even though its entropy is small. Concentrated beliefs help exactly when they are aimed correctly, and a confidently wrong belief is worse than an uninformed one. The MaaSSim belief-source decomposition (Table 14) exhibits the empirical counterpart, where the shuffled belief, sharp but attached to the wrong drivers, falls below the uniform prior.*

B.6 Rate Separation

Theorem B.18 (Rate Separation from Type-Agnostic Sampling). *There exists an HT-MG instance satisfying Assumptions B.1–B.6 on which PACT attains $\tilde{O}(\sqrt{K})$ regret while PSRL-NoType, which samples $\hat{\theta} \sim \text{Unif}(\Theta)$ each episode without updating, attains $\Omega(K)$ regret. The gap quantifies the value of online persona inference relative to type-agnostic sampling.*

Remark B.19 (Scope). *We do not claim a rate separation against an updating MAP estimator. MAP-Type-Greedy can fail on instances requiring exploration by the standard PSRL-vs-greedy gap (Osband, Russo, and Van Roy 2013), but that comparison is not part of the retained empirical claim set.*

B.7 Idealized PACT⁺ Burn-in Improvement

The *deep-trap* family $\mathcal{G}_m^{\text{trap}}$ is parameterized by (m, n) : $|\Theta_i| = m$ types per agent, n agents, $S = m + 2$ states, two actions per agent, deterministic unanimity-vote transitions, a distinguished jackpot type θ_{jack} with $r_i = 1$ at the rightmost state s^* and $r_i = 0.5$ elsewhere, uniform prior, true types $\theta_i^* = \theta_{\text{jack}}$ for all i .

Theorem B.20 (Regret of idealized PACT⁺). *Let $\beta > 0$. Under Assumptions B.1–B.4:*

(i) **Asymptotic match.** *For the pairwise-disagreement bonus in Eq. (2), the bound (6) carries over up to a K -independent additive constant $O(\beta n |\Theta_i| / \rho)$, since that bonus decays geometrically as the posterior concentrates. This statement does not apply to the nonvanishing action-spread term in the empirical proxy.*

(ii) **Burn-in separation.** *On $\mathcal{G}_m^{\text{trap}}$, expected burn-in regret is $\Theta(m^n H)$ for PACT and Joint-PSRL versus $O(H)$ for PACT⁺ at any $\beta > 0$.*

Proof sketch. (i) The PACT⁺-vs-PACT excess regret is bounded by the expected cumulative bonus, which under (TID) decays geometrically as $|\Theta_i| \exp(-\rho H k / 2)$ (Proposition B.16; Ghosal and Van der Vaart 2017). Summing gives the $O(\beta n |\Theta_i| / \rho)$ constant. (ii) PACT samples the all-jackpot profile with probability m^{-n} per episode. Off the jackpot state the marginal does not update (constant likelihood), giving expected wait m^n at $\Theta(H)$ regret each. PACT⁺'s bonus $D_k(s^*, a) = n(m - 1)/m^2 > 0$ at s^* back-propagates to

make the welfare gap strictly positive at every state regardless of the sampled profile. The exhaustive welfare-argmax planner steers to s^* in $S - 1$ steps and the marginal collapses in episode one. $\square$

At finite K , cumulative regret is capped at KH . The theorem is a property of the constructed deep-trap family and idealized bonus; the reported live scaling figures evaluate a different proxy and are not presented as a realization of this asymptotic separation.

B.8 Worst-Case Burn-in Without Reward Locality

Theorem B.21 (Exponential burn-in without RL). *For every n, m and $K \geq 1$, there is an HT-MG satisfying TI, PF, ISI, and DP but violating RL, with $|\Theta_i| = m$ and $A_{\text{joint}} = m^n$, on which every algorithm has Bayesian regret*

$$\mathbb{E}[\text{Reg}_i^B(K)] = \Omega(\min\{K, m^n\}). \quad (7)$$

In particular, at $K = \Theta(m^n)$ the expected burn-in is $\Omega(m^n)$.

Proof sketch. Let $M = m^n$, take $S = H = 1$, and index the M joint actions by the M joint type profiles. Draw θ^* from the product-uniform prior, which satisfies PF, and give every agent the deterministic reward $r_i(a, \theta) = \mathbf{1}[a = a^*(\theta)]$ for a bijection $a^*: \Theta \rightarrow A_{\text{joint}}$. The oracle chooses $a^*(\theta^*)$ and receives one each episode. Before the first reward of one, an algorithm can eliminate at most one candidate profile per distinct zero-reward action. Under the uniform prior, the position of the rewarding action in any adaptive no-repeat search order is uniform on $\{1, \dots, M\}$. Hence the expected number of zero-reward episodes is $\mathbb{E}[\min\{K, J - 1\}]$, where $J \sim \text{Unif}\{1, \dots, M\}$. This quantity is at least $K/2$ for $K \leq M/2$ and at least $(M - 1)/4$ for $K \geq M/2$, proving $\Omega(\min\{K, M\})$. Each r_i depends on the full joint type, so RL is violated. $\square$

Remark B.22 (Scope and what is open). *This construction keeps PF and drops RL; it therefore does not establish that PF is necessary. It also sets $A_{\text{joint}} = m^n$, so the exponential burn-in combines a joint-type reward with an exponentially large action menu and does not isolate persona-posterior storage from planning complexity. Whether an exponential decision-regret gap persists under RL, or with a fixed-size action menu, is open. The RL-only diagnostic of Figure 15 produces measurable posterior coupling but no resolved regret gap on its finite geometry. The positive decoupling result is therefore a sufficient representation theorem, not a tight tractability characterization.*

B.9 Empirical Validation

Table 4 shows that well-specified Joint-PSRL-Uniform and PACT remain unresolved under the same iid DGP, while Table 8 reports the complementary coupled-DGP robustness result; neither table is presented as a realization of the without-RL counterexample.

C Cost, Compute, and Prompts

This section reports the LLM-call budget, wall-clock cost, hardware/software environment, and prompt templates used by the LLM-facing components. Numbers are aggregated across the four backbones (DeepSeek-V3.2, GPT-5.4-nano, Kimi-K2.6, Llama-4-Maverick-17B-128E-Instruct-FP8) and the three substrates (HP-SPGG, Concordia, SOTOPIA-Hard).

C.1 LLM-call accounting

Per-episode LLM calls. The online numeric tracker issues no LLM calls: HP-SPGG evaluates its Gaussian likelihood analytically after one offline tensor calibration. Prompted HP-SPGG baselines issue agent/action calls as specified by their adapters. Compact Concordia uses an analytic payoff function and adds no judge call, whereas open-text SOTOPIA-Hard uses both action generations and a scenario judge. The practical bonus is computed from cached tensors and adds no provider calls.

Total-call scope. The historical experiments were produced by several runners whose retry logs and discarded attempts were not normalized into one billing ledger. We therefore do not infer a paper-wide provider-call total from episode counts. The auditable fresh E-A accepted-response counts are reported below; they exclude retries and should not be interpreted as billed calls.

Supplemental-run accounting. The historical E-A source audit and the E-B/E-D simulations make zero online model calls; E-D adds 324 live DeepSeek calibration calls. E-E also makes zero provider calls: it regenerates 50 MaaSSim sub-fleet cells and produces 240 tracker-level replay rows from analytic hidden-rule events. E-F adds 20 zero-provider bonus-ablation rows on saved states. The fresh environment-matched E-A control separately records 23,178 accepted response-cache entries: 6,000 complete calibration judge cells, 1,208 LLM-PSRL verbal entries, and 15,970 external-baseline entries. E-C adds 240 episodes for the requested $p=0.25, 0.5$ corruption cells and 240 corrected component-control episodes. Each accepted E-C episode records six agent generations and one judge call, giving 3,360 accepted-output calls, or 3,684 including E-D calibration. These accepted-output totals exclude transient provider retries; rejected fallback-bearing E-C attempts are checkpointed separately.

E-A replay and provider constraints. The four full-grid E-A calibrations each contain exactly 1,500 accepted cells and zero parse failures. External malformed outputs were accepted only after 262 cache-pinned strict JSON format repairs that preserved the numeric decisions; no heuristic parse fallback is permitted. Provider sampling seeds are unavailable, so environment seeds and type profiles are matched while raw accepted responses are replayed from SHA-256-keyed caches rather than claimed to be provider-RNG matched. The hosted Kimi endpoint rejects custom temperature, so its calibration and action requests used the supported value 1. The final LLM-PSRL belief update is skipped only after episode K , because no subsequent decision consumes it. Every tensor, cache, report, and aggregate, plus all 400 method/backbone/seed

NPZs, is covered by the release manifest together with tensor hashes, snapshot IDs, and UTC timestamps.

Artifact scope. The fresh E-A matched package is complete at the accepted-response and per-seed levels. Some earlier scaling NPZs and the historical SOTOPIA all-70 episode JSONs were removed during workspace cleanup; their retained figures and aggregate reports are not substitutes for raw trajectories. Reproducing those legacy cells therefore requires regenerating provider outputs, and provider RNG paths cannot be recreated exactly. The checklist marks data/code/seed availability as partial accordingly.

Token and monetary cost. The retained caches do not store authoritative provider token-usage or billing fields for every historical call, and the hosted Llama endpoint has no comparable public price. We therefore report accepted-response counts and wall-clock environment details rather than extrapolating a total token or dollar bill.

C.2 Wall-clock cost

Wall-clock time is dominated by provider latency and deployment-specific rate limits. Runs used bounded concurrency between one and five requests depending on the deployment; Kimi calls were serialized when required by its rate limit. Because historical terminal-level timing was not retained in a normalized per-cell ledger, we do not claim comparable median episode times across substrates.

C.3 Compute environment

Workstation. All orchestration, posterior updates, finite-action planner calls, and figure generation ran on a single workstation with an Intel x64 CPU, 64 GB RAM, and Windows 10 + PowerShell 5.1. No local GPU was used. LLM inference is offloaded entirely to the backbone providers' APIs. The Python environment is managed with `uv`, with separate virtual environments for the three substrates (HP-SPGG, Concordia, SOTOPIA-Hard) and a shared package layout under `external/`.

API backbones. The four backbones are accessed through an organization-managed provider proxy with token-based authentication. Per-call timeouts are 180 s with 2–3 retries; concurrency is deployment-specific and Kimi requests are serialized when required. Streaming is disabled. The models are pinned to DeepSeek-V3.2, GPT-5.4-nano-20260317, Kimi-K2.6, and Llama-4-Maverick-17B-128E-Instruct-FP8. The proxy implementation and credentials are not part of the anonymous artifact, so exact provider-path replay is unavailable.

Algorithmic compute. HP-SPGG uses backward induction with deterministic exhaustive joint-action argmax, and compact Concordia enumerates its one-step finite action menu. SOTOPIA-Hard has no finite planner. These finite searches run in < 1 ms per decision on the workstation CPU and are dominated by LLM latency. Posterior updates use closed-form Dirichlet conjugacy for the transition belief and direct Bayes' rule for the per-agent persona belief, both $O(n|\Theta_i|H)$ per episode and negligible relative to the LLM cost.

C.4 Prompt Templates

This appendix reports the prompt templates used by the LLM-facing components of PACT and PACT⁺. Templates are written with symbolic placeholders in braces. In the implementation, structured payloads are rendered as JSON and the model is instructed to return machine-parseable JSON when a structured response is required. For display in this two-column format, lines that exceed the column width are soft-wrapped ($\leftarrow$). The rendered prompt sent to the model contains no wrap marker.

HP-SPGG Persona-Conditioned Outcome Scoring
PACT uses persona-conditioned scores to build the reward/likelihood table for the controlled HP-SPGG substrate. The judge sees one candidate persona and one realized contribution profile, then returns a scalar satisfaction score.

```
[System]
Score the player satisfaction in a public-goods-game outcome. Return only valid JSON with key "score", a number from 0.0 to 1.0.

[User]
You are scoring exactly one HP-SPGG outcome. Your entire response must be one valid JSON object and nothing else. No explanation, no markdown, no prose. Use exactly this schema: {"score": 0.73}.

Persona key:           {persona_key}
Persona description: {persona_description}
Player index:         {player_index}
Player contribution: {own_contribution}
Other contributions: {other_contributions}
Group mean:           {group_mean_contribution}
Sample index:         {sample_index}

Score satisfaction from 0.0 to 1.0 using the persona's private behavioural profile. Return only the JSON object now.
```

PACT and PACT⁺ Action Selection in Open-Ended Tasks

For SOTOPIA-style open-ended interaction, PACT and PACT⁺ expose the maintained numeric posterior to the action-selection model. PACT uses the same template with no exploration bonus. PACT⁺ additionally sets $\beta = 0.25$ and the beta policy. The `baseline_instruction` and `beta_policy` fields are the only PACT-versus-PACT⁺ differences. All other fields are shared.

```
[System]
You are a careful social-interaction agent.
Return valid JSON only.

[User: JSON payload]
{
  "task": "SOTOPIA action selection",
  "agent": "{agent_name}",
  "baseline": "{pact_or_pact_plus}",
  "strategy_profile": "{strategy_profile}",
  "goal": "{private_goal}",
  "private_profile": "{private_profile_summary}",
  "observation": "{current_observation}",
  "history": "{recent_dialogue_history}",
```

```
"available_actions": {available_actions},
"action_instruction":
  "{environment_action_instruction}",

"baseline_instruction":
  "Maintain a posterior over the partner's hidden goal, type, and willingness to cooperate. Use the provided numeric persona posterior as the belief state before selecting a robust joint-value action. PACT: no exploration bonus. PACT+: add a small bonus only when a clarifying probe reduces uncertainty without sacrificing the current goal.",

"opponent_persona_posterior":
  {opponent_persona_posterior},
"opponent_persona_labels":
  {opponent_persona_labels},
"posterior_usage":
  "Treat as quantitative belief over partner personas. Prefer actions robust under the high-probability personas.",

"exploration_beta": 0.25,
"beta_policy":
  "Use exploratory probes only when information gain is useful and the probe still advances the current social goal.",

"response_schema": {
  "action_type": "one of available_actions",
  "argument": "utterance or content; empty if none",
  "to": "list of private recipients, usually
 $\leftarrow$  []",
  "reason": "brief reason"
},
"instruction":
  "Return only JSON. Use an action_type from available_actions."
}
```

Reproducibility Checklist

Instructions for Authors:

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are included} {(yes/partial/no)}  
yes
```

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1. General Paper Structure

- 1.1. Includes a conceptual outline and/or pseudocode description of AI methods introduced (yes/partial/no/NA) **yes**
- 1.2. Clearly delineates statements that are opinions, hypothesis, and speculation from objective facts and results (yes/no) **yes**
- 1.3. Provides well-marked pedagogical references for less-familiar readers to gain background necessary to replicate the paper (yes/no) **yes**

2. Theoretical Contributions

- 2.1. Does this paper make theoretical contributions? (yes/no) **yes**

If yes, please address the following points:

- 2.2. All assumptions and restrictions are stated clearly and formally (yes/partial/no) **yes**
- 2.3. All novel claims are stated formally (e.g., in theorem statements) (yes/partial/no) **yes**

- 2.4. Proofs of all novel claims are included (yes/partial/no) **yes**
- 2.5. Proof sketches or intuitions are given for complex and/or novel results (yes/partial/no) **yes**
- 2.6. Appropriate citations to theoretical tools used are given (yes/partial/no) **yes**
- 2.7. All theoretical claims are demonstrated empirically to hold (yes/partial/no/NA) **partial**
- 2.8. All experimental code used to eliminate or disprove claims is included (yes/no/NA) **no**

3. Dataset Usage

- 3.1. Does this paper rely on one or more datasets? (yes/no) **yes**

If yes, please address the following points:

- 3.2. A motivation is given for why the experiments are conducted on the selected datasets (yes/partial/no/NA) **yes**
- 3.3. All novel datasets introduced in this paper are included in a data appendix (yes/partial/no/NA) **partial**
- 3.4. All novel datasets introduced in this paper will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no/NA) **partial**
- 3.5. All datasets drawn from the existing literature (potentially including authors’ own previously published work) are accompanied by appropriate citations (yes/no/NA) **yes**
- 3.6. All datasets drawn from the existing literature (potentially including authors’ own previously published work) are publicly available (yes/partial/no/NA) **yes**
- 3.7. All datasets that are not publicly available are described in detail, with explanation why publicly available alternatives are not scientifically satisfying (yes/partial/no/NA) **NA**

4. Computational Experiments

- 4.1. Does this paper include computational experiments? (yes/no) **yes**

If yes, please address the following points:

- 4.2. This paper states the number and range of values tried per (hyper-) parameter during development of the paper, along with the criterion used for selecting the final parameter setting (yes/partial/no/NA) **partial**
- 4.3. Any code required for pre-processing data is included in the appendix (yes/partial/no) **no**
- 4.4. All source code required for conducting and analyzing

ing the experiments is included in a code appendix (yes/partial/no) [no](#)

- 4.5. All source code required for conducting and analyzing the experiments will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no) [yes](#)
- 4.6. All source code implementing new methods have comments detailing the implementation, with references to the paper where each step comes from (yes/partial/no) [partial](#)
- 4.7. If an algorithm depends on randomness, then the method used for setting seeds is described in a way sufficient to allow replication of results (yes/partial/no/NA) [partial](#)
- 4.8. This paper specifies the computing infrastructure used for running experiments (hardware and software), including GPU/CPU models; amount of memory; operating system; names and versions of relevant software libraries and frameworks (yes/partial/no) [partial](#)
- 4.9. This paper formally describes evaluation metrics used and explains the motivation for choosing these metrics (yes/partial/no) [yes](#)
- 4.10. This paper states the number of algorithm runs used to compute each reported result (yes/no) [yes](#)
- 4.11. Analysis of experiments goes beyond single-dimensional summaries of performance (e.g., average; median) to include measures of variation, confidence, or other distributional information (yes/no) [yes](#)
- 4.12. The significance of any improvement or decrease in performance is judged using appropriate statistical tests (e.g., Wilcoxon signed-rank) (yes/partial/no) [no](#)
- 4.13. This paper lists all final (hyper-)parameters used for each model/algorithm in the paper's experiments (yes/partial/no/NA) [partial](#)